Snowflake Milestone 2: MCQ Bank, Topic-wise

All **400 questions** from `MS2_SNOWFLAKE_MCQ_QUESTION__BANK.docx`, regrouped by the topic bifurcation in your screenshot. Nothing removed, nothing duplicated; every question appears once under exactly one topic.

How to find a question in the original file: `S1-Q12` = Set 1, question 12 (first 200 questions). `S2-Q12` = Set 2, question 12 (second half of the file, where numbering restarts at Q1). Difficulty is shown as E / M / H.

1. Coverage map (no topic missed)

Code	Topic (from your bifurcation)	Exam row	Exam MCQs x marks	Bank Qs	Set 1	Set 2
T1	Stored Procedures; Snowsight Modern UI; Monitoring Usage	Easy	4 x 1	38	20	18
T2	Snowflake Architecture: Storage, Compute, User Roles	Easy	4 x 1	79	48	31
T3	Supported Clouds; Joins	Easy	2 x 1	23	23	0
T4	Snowflake SQL: DDL, DML	Easy	4 x 1	36	36	0
T5	Visualizations (Matplotlib/Seaborn); SQL with Snowpark	Easy	4 x 1	27	0	27
T6	Snowflake Streamlit	Easy	2 x 1	17	0	17
T7	Snowflake Horizon	Medium	2 x 2	8	0	8
T8	Monitoring Usage; Dynamic Tables	Medium	3 x 2	29	8	21
T9	Snowflake Streams; Tasks; Iceberg Tables	Medium	1 x 2	19	6	13
T10	Snowpipe; Snowflake Integration with dbt	Difficult	2 x 2	24	3	21

Code	Topic (from your bifurcation)	Exam row	Exam MCQs x marks	Bank Qs	Set 1	Set 2
T11	Snowflake UDFs	Difficult	2 x 2	19	0	19
X1	Tables, Time Travel, Data Load & Unload, Sharing	(Coding section topic)	feeds Coding sections	56	50	6
X2	Semi-structured Data / Cortex	(Coding section topic)	feeds Coding sections	25	6	19
	Total		**30 MCQs = 40 marks**	**400**	**200**	**200**

- **T1 to T11** are the 11 MCQ rows of the bifurcation. **X1 and X2** are topics from the *Coding* rows (Tables/Time Travel/Load & Unload/Sharing, and Snowpark Semi-structured/Cortex) that the MCQ bank also tests, so they are separate sections, not dropped.
- **"Monitoring Usage" appears twice** in the bifurcation (Easy row with Stored Procedures/Snowsight, Medium row with Dynamic Tables). I split monitoring questions by difficulty: **Easy goes to T1, Medium/Hard goes to T8**.
- Warehouse settings (auto-suspend, auto-resume, sizing, multi-cluster) are filed under **T2 Compute**. Replication, regions and secondary databases are filed under **T3 Supported Clouds**.
- Some topics only exist in one set: **T3, T4 are Set 1 only; T5, T6, T7, T11 are Set 2 only.** Do both sets or you will miss topics.

2. How to remember 400 answers (the method)

1. You are not memorizing 400 things. Set 1's 200 questions are built from only **80 distinct answers** (the same fact is re-worded 2 to 3 times, e.g. *Secure Data Sharing* answers three different questions). Set 2's 200 questions have 167 distinct answers, and only 6 answers appear in both sets. In total that is about **240 distinct answers**, and many are the same idea in different words. Each topic below starts with a **Memory hooks** box and an **Answer sheet** that lists each distinct answer once, with the wording that triggers it.

2. Learn trigger phrase to answer, not whole questions. The exam rewords the story but keeps the trigger ("changed rows" = Stream, "schedule" = Task, "no copy" = Secure Data Sharing). Read the *Trigger* column and say the answer out loud.

3. Learn confusable pairs together (table below). Most wrong answers on this bank come from mixing up neighbours: Stream vs Task vs Dynamic Table, UDF vs Stored Procedure, scale up vs scale out, Time Travel vs Cloning vs Sharing.

4. Test by recall, not re-reading. Cover the answer, say it, then check. Use the flashcard CSV (`MS2_MCQ_Flashcards.csv`, one card per distinct answer) in Anki with spaced repetition: review on day 1, 2, 4, 7 and 14. Only the cards you miss keep coming back.

5. Don't use answer position. In both sets the correct answer is A, B, C and D exactly 50 times each, so position tells you nothing.

6. Study in exam-weight order. 20 of the 30 exam MCQs are 1-mark Easy ones (T1 to T6), so those topics are worth half the MCQ marks. Suggested 5 days:

Day	Do this	Why
1	T2, T3, T4 (Architecture, Clouds/Joins, SQL)	10 exam MCQs, mostly Set 1 facts
2	T1, T5, T6 (Procedures/Snowsight/Monitoring, Snowpark/Visualization, Streamlit)	10 exam MCQs
3	T7, T8, T9 (Horizon, Dynamic Tables/Monitoring, Streams/Tasks/Iceberg)	6 Medium MCQs at 2 marks = 12 marks
4	T10, T11, then X1, X2	4 Difficult MCQs, and X1/X2 double as coding-section revision
5	Mixed flashcards + redo only the questions you missed	Fix weak spots

Confusable pairs: the decision table

If the question says...	Answer	Don't mix it up with
Live data to another account, **no copy**	Secure Data Sharing	Zero-copy cloning (a copy inside your account)
Query or restore data as it was earlier	Time Travel (`AT or BEFORE`)	Cloning
Quick dev/test copy, little storage	Zero-copy cloning	Secure Data Sharing
Only changed rows since last run	Stream	Task (that's *when*, not *what changed*)

If the question says...	Answer	Don't mix it up with
Run on a schedule / CRON / in order	Task	Stream
Derived table stays fresh automatically, minimal orchestration	Dynamic Table	Streams+Tasks (explicit change capture + custom logic)
Files land in cloud storage, load them automatically	Snowpipe (event notifications; costs compute)	Manual COPY INTO
Several SQL steps, branching, parameters	Stored procedure (`CALL`)	UDF
Reusable calculation inside a SELECT	UDF (scalar = one value per call)	Stored procedure
Many users at once, queries queuing	Multi-cluster warehouse (scale OUT)	Bigger warehouse (scale UP)
One heavy query is slow	Larger warehouse size	Multi-cluster
ETL and BI compete for compute	Separate virtual warehouses	One shared warehouse
Same query, unchanged data, instant answer	Result cache	Clustering
Skip micro-partitions that can't match	Pruning (help it with a clustering key on large tables)	Result cache
Who accessed what, governance, compliance	Snowflake Horizon	Query Profile
Why is this query slow / expensive	Query Profile	Horizon
What did compute cost over time	`WAREHOUSE_METERING_HISTORY` / credits	Query History (what ran)
Evolving JSON	VARIANT	Fixed relational columns

If the question says...	Answer	Don't mix it up with
Semantic search over text	Cortex Search	Cortex Agents (natural-language assistant)
Custom Python model	Snowpark ML	Built-in AI functions
Static chart / stats plots on top of it	Matplotlib / Seaborn (built on Matplotlib)	—
Compare categories / relation of two numbers	Bar chart / Scatter plot	—
dbt dependency order	DAG	—
Open table format / lakehouse	Iceberg tables	—
Highest role / warehouses & DBs / given to everyone	ACCOUNTADMIN / SYSADMIN / PUBLIC	—
Test before loading / reload loaded file / stop-continue-skip	`VALIDATION_MODE='RETURN_ERRORS'` / `FORCE=TRUE` / `ON_ERROR`	—
Delete all rows, keep table / replace whole table	TRUNCATE TABLE / CREATE OR REPLACE TABLE	—

T1. Stored Procedures; Snowsight Modern UI; Monitoring Usage

Exam: 4 MCQ(s) x 1 mark(s), Easy row. Bank: ***38 questions*** *(Set 1: 20, Set 2: 18; Easy 27, Medium 9, Hard 2).*

Memory hooks

- **Stored procedure = a *process*** (several SQL steps, conditions, parameters). Create it with `CREATE PROCEDURE ... RETURNS ...`, run it with `CALL`.
- JavaScript procedures: `snowflake.execute()` runs SQL, **TRY / CATCH** handles errors, `while (rs.next())` walks the result rows.

- **Snowsight** = the browser UI (worksheets, **Dashboards**, **Query History / History tab**).
- Monitoring in one line: **History = what ran, Profile = how it ran, Metering = what it cost** (Query History / Query Profile / `WAREHOUSE_METERING_HISTORY`, credits consumed).

Answer sheet (29 distinct answers, most-asked first)

Answer	Asked	Trigger (what the question is really asking)
snowflake.execute()	2	execute SQL from a JavaScript stored procedure / submit a SQL statement from procedure code
TRY / CATCH	2	capture execution errors and implement controlled handling logic / handle JavaScript exceptions in a stored procedure
CALL procedure_name()	2	invoke an existing Snowflake stored procedure / execute a stored procedure
RETURNS	2	specify the data type returned by procedure execution / declare the return type of a stored procedure
CREATE PROCEDURE	2	create reusable procedural logic that can be invoked later / define a stored procedure
while (rs.next())	2	process each returned row from a statement result set / iterate through a JavaScript result set row by row
Query Profile	2	visualize query execution operators and performance metrics / Which Snowflake feature visually breaks down how a SQL statement spent time during execution?
Stored procedure	2	A Snowflake routine needs to run several SQL statements as one reusable unit. Which object is designed for this? / A finance workflow must run several SQL statements, use conditions, and return a status message. Which object best matches the requirement?

Answer	Asked	Trigger (what the question is really asking)
Query History	2	Which history area is most useful for locating a query that ran much longer than expected? / Which Snowflake history source is relevant when reviewing executed SQL statements over time?
Execution Graph	1	identify expensive operators such as joins, scans, and aggregations in Query Profile
QUERY_HISTORY	1	inspect metadata about previously executed queries
Partitions Scanned	1	measure how many micro-partitions a query read
Resource Monitor	1	track warehouse credit use and apply thresholds, alerts, or suspension actions
History tab	1	review completed query activity in the Snowflake interface
Warehouse Usage History	1	review historical warehouse utilization and credit consumption
Bytes Scanned vs. Bytes Returned	1	compare data processed with data ultimately returned
CALL	1	Which SQL keyword is used when invoking a stored procedure that has already been created?
Input parameters	1	Which capability allows a Snowflake stored procedure to receive values from the caller?
Stored procedure logic	1	Which construct is most appropriate when a routine must branch based on conditions?
Snowsight	1	Which Snowflake interface provides browser-based worksheets for writing and running SQL?

Answer	Asked	Trigger (what the question is really asking)
Dashboards	1	A trainer wants to show saved visual tiles based on SQL results in Snowflake. Which Snowsight feature is relevant?
Execution duration	1	Which metric is directly useful when evaluating whether a query is taking too long?
WAREHOUSE_METERING_HISTORY	1	Which Account Usage view family is designed to expose historical warehouse credit consumption?
Credits consumed	1	Which measure is most directly tied to Snowflake compute cost decisions?
To balance performance and compute cost	1	Why would a team compare query runtime with warehouse size?
Call a parameterized stored procedure	1	A Snowflake routine should be invoked with a customer_id and process multiple SQL steps based on that value. Which invocation model is appr…
Parameterize the procedure	1	A stored procedure must run a set of business rules repeatedly for different regions. What design improves reusability?
Define clear parameters and return behavior	1	A stored procedure must support reusable business logic across many callers while preserving consistent input handling. Which design choice…
Snowsight + Snowpark + Streamlit + Cortex capabilities	1	A user wants a single platform for SQL dashboards, Python transformations, interactive apps, and AI-assisted data interaction. Which combin…

All 38 questions with answers

- **S1-Q13** (E) For the requirement "execute SQL from a JavaScript stored procedure," which Snowflake term is the best fit? -> **snowflake.execute()**. snowflake.execute() is used to

execute SQL in JavaScript stored procedures.

- **S1-Q25** (E) Which option best matches this Snowflake responsibility: identify expensive operators such as joins, scans, and aggregations in Query Profile? -> **Execution Graph**. The execution graph shows operators and their relationships in a query plan.
- **S1-Q27** (M) A logistics platform is being redesigned so it can capture execution errors and implement controlled handling logic. Which choice is most appropriate? -> **TRY / CATCH**. JavaScript TRY/CATCH provides exception handling in stored procedures.
- **S1-Q31** (M) The data team at a SaaS company must submit a SQL statement from procedure code. Which Snowflake feature or command should they select? -> **snowflake.execute()**. snowflake.execute() is used to execute SQL in JavaScript stored procedures.
- **S1-Q42** (M) A financial services team needs to invoke an existing Snowflake stored procedure. Which Snowflake option best meets this requirement? -> **CALL procedure_name()**. CALL is used to execute a stored procedure.
- **S1-Q51** (E) A trainee needs to identify the Snowflake feature that can inspect metadata about previously executed queries. Which choice is correct? -> **QUERY_HISTORY**. QUERY_HISTORY exposes query execution history metadata.
- **S1-Q64** (M) A logistics platform is being redesigned so it can specify the data type returned by procedure execution. Which choice is most appropriate? -> **RETURNS**. RETURNS declares the procedure return type.
- **S1-Q72** (M) An architect for a telecommunications system has one key requirement: create reusable procedural logic that can be invoked later. Which option best satisfies it? -> **CREATE PROCEDURE**. CREATE PROCEDURE defines a stored procedure.
- **S1-Q73** (E) Which Snowflake concept is primarily used to measure how many micro-partitions a query read? -> **Partitions Scanned**. Partitions Scanned indicates the micro-partitions read by a query.
- **S1-Q98** (E) Which Snowflake capability directly supports the need to execute a stored procedure? -> **CALL procedure_name()**. CALL is used to execute a stored procedure.
- **S1-Q117** (E) For the requirement "track warehouse credit use and apply thresholds, alerts, or suspension actions," which Snowflake term is the best fit? -> **Resource Monitor**. Resource monitors govern credit consumption and can trigger notifications or suspension actions.
- **S1-Q125** (E) Which Snowflake concept is primarily used to review completed query activity in the Snowflake interface? -> **History tab**. The History area provides access to completed query activity.
- **S1-Q141** (M) The data team at a education company must process each returned row from a statement result set. Which Snowflake feature or command should they select? -> **while (rs.next())**. Result sets are commonly advanced with rs.next() in JavaScript procedures.
- **S1-Q171** (E) A trainee needs to identify the Snowflake feature that can visualize query execution operators and performance metrics. Which choice is correct? -> **Query Profile**. Query Profile breaks down query execution and performance.

- **S1-Q174** (E) For the requirement "handle JavaScript exceptions in a stored procedure," which Snowflake term is the best fit? -> **TRY / CATCH**. JavaScript TRY/CATCH provides exception handling in stored procedures.
- **S1-Q181** (E) Which Snowflake capability directly supports the need to review historical warehouse utilization and credit consumption? -> **Warehouse Usage History**. Warehouse usage history provides historical warehouse activity and credit information.
- **S1-Q182** (E) For the requirement "compare data processed with data ultimately returned," which Snowflake term is the best fit? -> **Bytes Scanned vs. Bytes Returned**. Comparing bytes scanned with bytes returned can reveal inefficient scanning.
- **S1-Q183** (E) Which option best matches this Snowflake responsibility: declare the return type of a stored procedure? -> **RETURNS**. RETURNS declares the procedure return type.
- **S1-Q196** (E) Which Snowflake concept is primarily used to iterate through a JavaScript result set row by row? -> **while (rs.next())**. Result sets are commonly advanced with rs.next() in JavaScript procedures.
- **S1-Q198** (E) For the requirement "define a stored procedure," which Snowflake term is the best fit? -> **CREATE PROCEDURE**. CREATE PROCEDURE defines a stored procedure.
- **S2-Q1** (E) A Snowflake routine needs to run several SQL statements as one reusable unit. Which object is designed for this? -> **Stored procedure**. Stored procedures package multi-step procedural logic for reusable execution.
- **S2-Q2** (E) Which SQL keyword is used when invoking a stored procedure that has already been created? -> **CALL**. CALL is the Snowflake statement used to invoke a stored procedure.
- **S2-Q3** (E) Which capability allows a Snowflake stored procedure to receive values from the caller? -> **Input parameters**. Stored procedures can define parameters that are supplied when the procedure is called.
- **S2-Q4** (E) Which construct is most appropriate when a routine must branch based on conditions? -> **Stored procedure logic**. Procedural constructs such as IF logic belong in stored procedures rather than passive objects like stages or caches.
- **S2-Q16** (E) Which Snowflake interface provides browser-based worksheets for writing and running SQL? -> **Snowsight**. Snowsight provides interactive worksheets and related UI experiences.
- **S2-Q17** (E) A trainer wants to show saved visual tiles based on SQL results in Snowflake. Which Snowsight feature is relevant? -> **Dashboards**. Snowsight dashboards visualize query results in reusable views.
- **S2-Q18** (E) Which history area is most useful for locating a query that ran much longer than expected? -> **Query History**. Query History exposes execution details for completed queries.
- **S2-Q19** (E) Which metric is directly useful when evaluating whether a query is taking too long? -> **Execution duration**. Execution duration is a core performance metric for query tuning.
- **S2-Q20** (E) Which Account Usage view family is designed to expose historical warehouse credit consumption? -> **WAREHOUSE_METERING_HISTORY**.

WAREHOUSE_METERING_HISTORY records warehouse metering and credit usage.

- **S2-Q21** (E) Which measure is most directly tied to Snowflake compute cost decisions? -> **Credits consumed**. Snowflake compute consumption is measured in credits.
- **S2-Q22** (E) Why would a team compare query runtime with warehouse size? -> **To balance performance and compute cost**. Runtime versus warehouse size helps determine whether extra compute is justified.
- **S2-Q72** (E) Which Snowflake feature visually breaks down how a SQL statement spent time during execution? -> **Query Profile**. Query Profile helps analyze execution steps and performance characteristics.
- **S2-Q73** (E) Which Snowflake history source is relevant when reviewing executed SQL statements over time? -> **Query History**. Query History records information about executed queries.
- **S2-Q89** (M) A finance workflow must run several SQL statements, use conditions, and return a status message. Which object best matches the requirement? -> **Stored procedure**. Stored procedures support multi-step procedural logic and can return status values.
- **S2-Q90** (M) A Snowflake routine should be invoked with a customer_id and process multiple SQL steps based on that value. Which invocation model is appropriate? -> **Call a parameterized stored procedure**. Stored procedures can accept input parameters supplied by the caller.
- **S2-Q148** (M) A stored procedure must run a set of business rules repeatedly for different regions. What design improves reusability? -> **Parameterize the procedure**. Parameters allow one procedure to handle multiple input values.
- **S2-Q165** (H) A stored procedure must support reusable business logic across many callers while preserving consistent input handling. Which design choice is most important? -> **Define clear parameters and return behavior**. Parameterized procedures are reusable and easier to operate consistently.
- **S2-Q198** (H) A user wants a single platform for SQL dashboards, Python transformations, interactive apps, and AI-assisted data interaction. Which combination reflects the source scope? -> **Snowsight + Snowpark + Streamlit + Cortex capabilities**. The source spans these platform capabilities for analytics, engineering, apps, and AI.

T2. Snowflake Architecture: Storage, Compute, User Roles

*Exam: 4 MCQ(s) x 1 mark(s), Easy row. Bank: **79 questions** (Set 1: 48, Set 2: 31; Easy 30, Medium 32, Hard 17).*

Memory hooks

- **3 layers:** Storage (persistent data, micro-partitions) / Compute (virtual warehouses) /

Services (metadata, security, parsing, optimization). Compute and storage scale **independently**, so teams can share data with separate warehouses.

- **Roles:** ACCOUNTADMIN = highest level; SYSADMIN = warehouses & databases; PUBLIC = granted to everyone automatically; RBAC = privileges are given to roles, not users; role hierarchy = inheritance.
- **Scale UP vs OUT:** bigger warehouse = one heavy query; **multi-cluster** = many users at once (concurrency/queuing); **separate warehouses** = stop ETL and BI from fighting.
- **Auto-Suspend** stops an idle warehouse (saves credits); **Auto-Resume** starts it when a query arrives. Use both together.
- **Result cache** = same query + unchanged data returns instantly; it is **not reusable once underlying data changes**.
- **Pruning** = skipping micro-partitions using metadata. **Clustering key** only for very large tables with repeated selective filters (not every table); `ALTER TABLE ... RECLUSTER` is the manual command.

Answer sheet (46 distinct answers, most-asked first)

Answer	Asked	Trigger (what the question is really asking)
ACCOUNTADMIN	3	perform the highest-level account administration tasks / manage account-wide administrative capabilities when the highest system role is required
SYSADMIN	3	create and manage warehouses and databases under the standard system hierarchy / handle routine object and compute administration without defaulting to the highest role
Role-Based Access Control (RBAC)	3	manage privileges centrally by job function instead of user-by-user grants / assign privileges to roles and then assign roles to users
Services Layer	3	centralize metadata and coordinate query processing / manage metadata, authentication, query parsing, and optimization
Virtual Warehouse	3	provide compute resources for query processing / run SQL workloads without storing table data inside the compute resource
Storage Layer	3	persistent storage remains available independently of compute / persist table data in compressed columnar

Answer	Asked	Trigger (what the question is really asking)
		micro-partitions
Micro-partitions	3	organize table data into small columnar storage units with metadata / use min/max and other metadata to reduce data scanning
PUBLIC	3	grant only intentionally broad privileges that all users may inherit / provide privileges inherited by every user in the account
Independent Compute and Storage	3	let multiple workloads use the same stored data with separate compute / scale compute without duplicating the underlying stored data
Micro-Partition Pruning	3	avoid scanning irrelevant partitions for selective predicates / skip storage partitions that cannot satisfy query filters
Result cache	3	Which feature can return a previous query result without recomputing it when cache conditions remain valid? / A BI query runs repeatedly against unchanged data and returns much faster after the first execution. Which mechanism is most likely respons…
Clustering Key	2	improve micro-partition organization for columns used frequently in selective filters or joins / improve pruning on a very large table whose natural clustering has degraded
Auto-Resume	2	let users submit queries without manually resuming the warehouse first / restart a suspended warehouse when new work arrives
Result Cache	2	reuse an eligible prior query result without recomputing it / return an identical unchanged query result even if compute was later suspended
Larger Warehouse Size	2	increase compute resources for a single heavy workload / reduce elapsed time for a compute-

Answer	Asked	Trigger (what the question is really asking)
		intensive query when more per-query resources are needed
Multi-Cluster Warehouse	2	reduce query queuing when many users submit work simultaneously / add or remove clusters to handle changing query concurrency
ALTER TABLE ... RECLUSTER	2	manually invoke reclustering for a table in the context of the source question set / request reclustering of a table using the source-referenced syntax
Auto-Suspend	2	avoid paying compute credits while a warehouse is inactive / stop an idle virtual warehouse after a configured period
Separate Virtual Warehouses	2	isolate independent workloads while using the same stored data / prevent ETL and BI workloads from competing for the same compute
Appropriate Warehouse Sizing	2	balance performance and compute consumption for a workload / match compute size to workload requirements instead of using one size for every query
AUTO_RESUME	2	Which warehouse setting lets compute start automatically when a new query arrives? / A dashboard warehouse is suspended overnight but should restart automatically when users arrive in the morning. Which setting should be ena…
Clustering	2	Which technique can improve pruning when a very large table is repeatedly filtered on the same columns? / A large table has poor pruning on region and event_date after years of growth. Which technique can improve data organization for those filt…
AUTO_SUSPEND	1	Which configuration stops an idle warehouse automatically after a defined period?
RBAC	1	Which mechanism is the foundation for least-privilege access control in Snowflake?

Answer	Asked	Trigger (what the question is really asking)
Reuse privileges through a role hierarchy	1	What does role inheritance help administrators achieve?
Independent compute and storage	1	Which architectural principle allows multiple workloads to use the same stored data with different compute resources?
Services layer	1	Which Snowflake layer is associated with query parsing, optimization, and metadata services?
Partition pruning	1	Which concept helps Snowflake avoid reading micro-partitions that cannot satisfy a filter?
High query concurrency	1	What problem do multi-cluster warehouses mainly address?
Use separate virtual warehouses	1	For workload isolation, what is generally more effective than sending every team to one shared warehouse?
Configure a shorter AUTO_SUSPEND interval	1	A warehouse remains active for long idle gaps between hourly jobs. What change most directly reduces wasted compute?
Multi-cluster warehouse	1	A company needs many dashboard users to query concurrently during month-end. Queuing appears even though individual queries are not unusual…
Assign separate virtual warehouses	1	An ETL team and a reporting team access the same tables but should not compete for compute. Which design best matches Snowflake architectur…
Micro-partition pruning	1	A very large order table is frequently filtered on order_date. Snowflake should skip unrelated storage units. Which concept makes that poss…
Snowflake executes multiple workloads close to shared data	1	A platform wants AI dashboards, Python transformations, and SQL analytics to use the same governed data without exporting copies. Which arc…

Answer	Asked	Trigger (what the question is really asking)
Separate virtual warehouses per workload	1	A company has ETL, data science, and BI workloads with very different resource needs. Which design provides workload isolation?
Pruning and clustering effectiveness	1	A query reads many partitions but returns very few rows. Which optimization topic should the team investigate first?
Role-based access control	1	A business user should see a summarized chart, but not receive unrestricted access to underlying tables. Which principle should the solutio…
Underlying data changed	1	A dashboard query result is stale because source data changed. Why might the result cache no longer be reusable?
Scale out with a multi-cluster warehouse	1	A dashboard workload has 500 concurrent short queries. Scaling a single warehouse up improves individual-query speed but queuing remains. W…
Use separate virtual warehouses over the same data	1	ETL and BI teams use the same warehouse. During nightly processing, dashboard latency becomes unacceptable. What architecture change best i…
Pruning effectiveness and clustering strategy	1	A query scans many micro-partitions even though predicates are highly selective. Which pair of areas should be reviewed before simply scali…
Clustering should be targeted to tables and columns with demonstrated pruning benefit	1	A team adds a clustering key to every large table. Costs rise and some queries show little improvement. What is the best interpretation?
The cached result is no longer eligible because underlying data changed	1	A repeated report query was fast yesterday due to cache reuse but is slow today after source data was refreshed. What is the most plausible…
AUTO_SUSPEND with AUTO_RESUME	1	A company wants low-cost idle periods and rapid restart for bursty analytics. Which warehouse configuration pairing best fits?

	Answer	Asked	Trigger (what the question is really asking)
1	**Compute and storage are decoupled**		A company uses separate warehouses for ETL and BI but both still read the same data. Which Snowflake architecture principle makes this poss…

All 79 questions with answers

- **S1-Q3** (E) For the requirement "perform the highest-level account administration tasks," which Snowflake term is the best fit? -> **ACCOUNTADMIN**. ACCOUNTADMIN is Snowflake's highest-level system role and should be used carefully.
- **S1-Q5** (E) For the requirement "improve micro-partition organization for columns used frequently in selective filters or joins," which Snowflake term is the best fit? -> **Clustering Key**. Clustering keys can improve pruning for large tables with suitable access patterns.
- **S1-Q6** (E) Which option best matches this Snowflake responsibility: create and manage warehouses and databases under the standard system hierarchy? -> **SYSADMIN**. SYSADMIN is generally used for warehouses, databases, and related objects.
- **S1-Q8** (M) In a logistics workload, engineers want to manage privileges centrally by job function instead of user-by-user grants. What should they use? -> **Role-Based Access Control (RBAC)**. RBAC centralizes privilege management through roles.
- **S1-Q10** (M) An architect for a marketing system has one key requirement: let users submit queries without manually resuming the warehouse first. Which option best satisfies it? -> **Auto-Resume**. Auto-Resume starts compute when work is submitted.
- **S1-Q11** (H) A insurance platform must centralize metadata and coordinate query processing. The team wants the most direct Snowflake mechanism rather than a workaround. Which choice is best? -> **Services Layer**. The Services Layer manages metadata, security coordination, parsing, and optimization.
- **S1-Q14** (E) For the requirement "provide compute resources for query processing," which Snowflake term is the best fit? -> **Virtual Warehouse**. A virtual warehouse supplies compute for queries and data processing.
- **S1-Q24** (H) During a design review for a insurance solution, the team must ensure it can run SQL workloads without storing table data inside the compute resource. Which Snowflake option is the strongest architectural fit? -> **Virtual Warehouse**. A virtual warehouse supplies compute for queries and data processing.
- **S1-Q39** (H) During a design review for a digital payments solution, the team must ensure it can persistent storage remains available independently of compute. Which Snowflake option is the strongest architectural fit? -> **Storage Layer**. The Storage Layer persists table data and organizes it into compressed micro-partitions.
- **S1-Q46** (E) A trainee needs to identify the Snowflake feature that can reuse an eligible prior query result without recomputing it. Which choice is correct? -> **Result Cache**. The result

cache stores eligible query results independently of warehouse local cache.

- **S1-Q56** (E) A trainee needs to identify the Snowflake feature that can organize table data into small columnar storage units with metadata. Which choice is correct? -> **Micro-partitions**. Snowflake automatically stores table data in micro-partitions with metadata used for pruning.
- **S1-Q58** (M) A digital payments team needs to run SQL workloads without storing table data inside the compute resource. Which Snowflake option best meets this requirement? -> **Virtual Warehouse**. A virtual warehouse supplies compute for queries and data processing.
- **S1-Q59** (M) A SaaS platform is being redesigned so it can grant only intentionally broad privileges that all users may inherit. Which choice is most appropriate? -> **PUBLIC**. PUBLIC is automatically granted to all users, so privileges on it are broadly inherited.
- **S1-Q61** (E) A trainee needs to identify the Snowflake feature that can increase compute resources for a single heavy workload. Which choice is correct? -> **Larger Warehouse Size**. Resizing a warehouse changes compute capacity available to a workload.
- **S1-Q71** (M) The data team at a travel company must improve pruning on a very large table whose natural clustering has degraded. Which Snowflake feature or command should they select? -> **Clustering Key**. Clustering keys can improve pruning for large tables with suitable access patterns.
- **S1-Q79** (H) A retail engineer is asked to meet this constraint: let multiple workloads use the same stored data with separate compute. Which option addresses the constraint with the least unnecessary complexity? -> **Independent Compute and Storage**. Snowflake separates compute from storage so each can scale independently.
- **S1-Q88** (M) The data team at a marketing company must reduce query queuing when many users submit work simultaneously. Which Snowflake feature or command should they select? -> **Multi-Cluster Warehouse**. Multi-cluster warehouses scale out for concurrency.
- **S1-Q93** (E) Which Snowflake concept is primarily used to scale compute without duplicating the underlying stored data? -> **Independent Compute and Storage**. Snowflake separates compute from storage so each can scale independently.
- **S1-Q95** (M) The data team at a digital payments company must let multiple workloads use the same stored data with separate compute. Which Snowflake feature or command should they select? -> **Independent Compute and Storage**. Snowflake separates compute from storage so each can scale independently.
- **S1-Q97** (M) A e-commerce platform is being redesigned so it can manually invoke reclustering for a table in the context of the source question set. Which choice is most appropriate? -> **ALTER TABLE ... RECLUSTER**. The source assessment identifies ALTER TABLE ... RECLUSTER as the manual reclustering command.
- **S1-Q100** (M) An architect for a supply chain system has one key requirement: avoid paying compute credits while a warehouse is inactive. Which option best satisfies it? -> **Auto-Suspend**. Auto-Suspend stops idle warehouse compute.
- **S1-Q119** (E) Which Snowflake concept is primarily used to persist table data in compressed columnar micro-partitions? -> **Storage Layer**. The Storage Layer persists table data and

organizes it into compressed micro-partitions.

- **S1-Q122** (E) Which Snowflake concept is primarily used to isolate independent workloads while using the same stored data? -> **Separate Virtual Warehouses**. Separate warehouses provide workload isolation because compute is independent.
- **S1-Q124** (E) Which option best matches this Snowflake responsibility: stop an idle virtual warehouse after a configured period? -> **Auto-Suspend**. Auto-Suspend stops idle warehouse compute.
- **S1-Q130** (M) The data team at a media company must balance performance and compute consumption for a workload. Which Snowflake feature or command should they select? -> **Appropriate Warehouse Sizing**. Warehouse size should be appropriate to query complexity and performance goals.
- **S1-Q131** (H) A travel team is troubleshooting a design. The required outcome is to avoid scanning irrelevant partitions for selective predicates, but the current approach does not provide that behavior. Which option should replace the current approach? -> **Micro-Partition Pruning**. Pruning uses micro-partition metadata to eliminate irrelevant partitions from a scan.
- **S1-Q132** (H) A manufacturing engineer is asked to meet this constraint: use min/max and other metadata to reduce data scanning. Which option addresses the constraint with the least unnecessary complexity? -> **Micro-partitions**. Snowflake automatically stores table data in micro-partitions with metadata used for pruning.
- **S1-Q133** (H) A logistics platform must grant only intentionally broad privileges that all users may inherit. The team wants the most direct Snowflake mechanism rather than a workaround. Which choice is best? -> **PUBLIC**. PUBLIC is automatically granted to all users, so privileges on it are broadly inherited.
- **S1-Q134** (M) In a travel workload, engineers want to return an identical unchanged query result even if compute was later suspended. What should they use? -> **Result Cache**. The result cache stores eligible query results independently of warehouse local cache.
- **S1-Q135** (E) Which Snowflake capability directly supports the need to assign privileges to roles and then assign roles to users? -> **Role-Based Access Control (RBAC)**. RBAC centralizes privilege management through roles.
- **S1-Q138** (M) In a insurance workload, engineers want to use min/max and other metadata to reduce data scanning. What should they use? -> **Micro-partitions**. Snowflake automatically stores table data in micro-partitions with metadata used for pruning.
- **S1-Q139** (M) A insurance team needs to manage account-wide administrative capabilities when the highest system role is required. Which Snowflake option best meets this requirement? -> **ACCOUNTADMIN**. ACCOUNTADMIN is Snowflake's highest-level system role and should be used carefully.
- **S1-Q146** (H) During a design review for a logistics solution, the team must ensure it can manage privileges centrally by job function instead of user-by-user grants. Which Snowflake option is the strongest architectural fit? -> **Role-Based Access Control (RBAC)**. RBAC centralizes privilege management through roles.

- **S1-Q148** (M) A financial services team needs to avoid scanning irrelevant partitions for selective predicates. Which Snowflake option best meets this requirement? -> **Micro-Partition Pruning**. Pruning uses micro-partition metadata to eliminate irrelevant partitions from a scan.
- **S1-Q149** (M) An architect for a media system has one key requirement: persistent storage remains available independently of compute. Which option best satisfies it? -> **Storage Layer**. The Storage Layer persists table data and organizes it into compressed micro-partitions.
- **S1-Q150** (M) In a banking workload, engineers want to reduce elapsed time for a compute-intensive query when more per-query resources are needed. What should they use? -> **Larger Warehouse Size**. Resizing a warehouse changes compute capacity available to a workload.
- **S1-Q156** (E) Which option best matches this Snowflake responsibility: provide privileges inherited by every user in the account? -> **PUBLIC**. PUBLIC is automatically granted to all users, so privileges on it are broadly inherited.
- **S1-Q158** (E) Which Snowflake concept is primarily used to match compute size to workload requirements instead of using one size for every query? -> **Appropriate Warehouse Sizing**. Warehouse size should be appropriate to query complexity and performance goals.
- **S1-Q160** (H) A financial services team is troubleshooting a design. The required outcome is to manage account-wide administrative capabilities when the highest system role is required, but the current approach does not provide that behavior. Which option should replace the current approach? -> **ACCOUNTADMIN**. ACCOUNTADMIN is Snowflake's highest-level system role and should be used carefully.
- **S1-Q169** (E) For the requirement "restart a suspended warehouse when new work arrives," which Snowflake term is the best fit? -> **Auto-Resume**. Auto-Resume starts compute when work is submitted.
- **S1-Q175** (H) A financial services platform must handle routine object and compute administration without defaulting to the highest role. The team wants the most direct Snowflake mechanism rather than a workaround. Which choice is best? -> **SYSADMIN**. SYSADMIN is generally used for warehouses, databases, and related objects.
- **S1-Q176** (M) A education platform is being redesigned so it can handle routine object and compute administration without defaulting to the highest role. Which choice is most appropriate? -> **SYSADMIN**. SYSADMIN is generally used for warehouses, databases, and related objects.
- **S1-Q177** (E) Which option best matches this Snowflake responsibility: manage metadata, authentication, query parsing, and optimization? -> **Services Layer**. The Services Layer manages metadata, security coordination, parsing, and optimization.
- **S1-Q180** (E) A trainee needs to identify the Snowflake feature that can add or remove clusters to handle changing query concurrency. Which choice is correct? -> **Multi-Cluster Warehouse**. Multi-cluster warehouses scale out for concurrency.
- **S1-Q188** (M) In a e-commerce workload, engineers want to centralize metadata and coordinate query processing. What should they use? -> **Services Layer**. The Services Layer

manages metadata, security coordination, parsing, and optimization.

- **S1-Q193** (M) In a e-commerce workload, engineers want to prevent ETL and BI workloads from competing for the same compute. What should they use? -> **Separate Virtual Warehouses**. Separate warehouses provide workload isolation because compute is independent.
- **S1-Q194** (E) Which option best matches this Snowflake responsibility: request reclustering of a table using the source-referenced syntax? -> **ALTER TABLE ... RECLUSTER**. The source assessment identifies ALTER TABLE ... RECLUSTER as the manual reclustering command.
- **S1-Q199** (E) Which Snowflake capability directly supports the need to skip storage partitions that cannot satisfy query filters? -> **Micro-Partition Pruning**. Pruning uses micro-partition metadata to eliminate irrelevant partitions from a scan.
- **S2-Q23** (E) Which configuration stops an idle warehouse automatically after a defined period? -> **AUTO_SUSPEND**. AUTO_SUSPEND stops idle compute after the configured timeout.
- **S2-Q24** (E) Which warehouse setting lets compute start automatically when a new query arrives? -> **AUTO_RESUME**. AUTO_RESUME restarts suspended compute when work is submitted.
- **S2-Q55** (E) Which mechanism is the foundation for least-privilege access control in Snowflake? -> **RBAC**. Role-based access control assigns privileges through roles.
- **S2-Q56** (E) What does role inheritance help administrators achieve? -> **Reuse privileges through a role hierarchy**. Role inheritance allows higher-level roles to inherit privileges from lower-level roles.
- **S2-Q62** (E) Which architectural principle allows multiple workloads to use the same stored data with different compute resources? -> **Independent compute and storage**. Snowflake separates compute from storage so workloads can scale independently.
- **S2-Q63** (E) Which Snowflake layer is associated with query parsing, optimization, and metadata services? -> **Services layer**. The services layer coordinates metadata and query planning functions.
- **S2-Q64** (E) Which feature can return a previous query result without recomputing it when cache conditions remain valid? -> **Result cache**. Result cache can serve previously computed results when eligible.
- **S2-Q68** (E) Which concept helps Snowflake avoid reading micro-partitions that cannot satisfy a filter? -> **Partition pruning**. Micro-partition metadata enables pruning of irrelevant storage units.
- **S2-Q69** (E) Which technique can improve pruning when a very large table is repeatedly filtered on the same columns? -> **Clustering**. Clustering can improve data organization for frequently filtered columns.
- **S2-Q70** (E) What problem do multi-cluster warehouses mainly address? -> **High query concurrency**. Multi-cluster warehouses scale out to handle concurrent query demand.
- **S2-Q71** (E) For workload isolation, what is generally more effective than sending every team to one shared warehouse? -> **Use separate virtual warehouses**. Separate warehouses isolate

compute contention while sharing the same data.

- **S2-Q94** (M) A warehouse remains active for long idle gaps between hourly jobs. What change most directly reduces wasted compute? -> **Configure a shorter AUTO_SUSPEND interval**. AUTO_SUSPEND stops idle compute and therefore reduces unnecessary credit usage.
- **S2-Q95** (M) A dashboard warehouse is suspended overnight but should restart automatically when users arrive in the morning. Which setting should be enabled? -> **AUTO_RESUME**. AUTO_RESUME starts a suspended warehouse when new work is submitted.
- **S2-Q96** (M) A company needs many dashboard users to query concurrently during month-end. Queuing appears even though individual queries are not unusually large. Which capability best addresses concurrency? -> **Multi-cluster warehouse**. Multi-cluster warehouses scale out when concurrency rises.
- **S2-Q97** (M) An ETL team and a reporting team access the same tables but should not compete for compute. Which design best matches Snowflake architecture? -> **Assign separate virtual warehouses**. Separate virtual warehouses isolate compute while keeping shared storage.
- **S2-Q130** (M) A BI query runs repeatedly against unchanged data and returns much faster after the first execution. Which mechanism is most likely responsible? -> **Result cache**. The result cache can reuse eligible query results when underlying data has not changed.
- **S2-Q135** (M) A very large order table is frequently filtered on order_date. Snowflake should skip unrelated storage units. Which concept makes that possible? -> **Micro-partition pruning**. Pruning uses micro-partition metadata to avoid scanning irrelevant partitions.
- **S2-Q136** (M) A large table has poor pruning on region and event_date after years of growth. Which technique can improve data organization for those filters? -> **Clustering**. Clustering can improve pruning for frequently filtered columns.
- **S2-Q137** (M) A platform wants AI dashboards, Python transformations, and SQL analytics to use the same governed data without exporting copies. Which architectural advantage enables this? -> **Snowflake executes multiple workloads close to shared data**. Snowflake allows diverse workloads to operate on shared governed data with separate compute.
- **S2-Q138** (M) A company has ETL, data science, and BI workloads with very different resource needs. Which design provides workload isolation? -> **Separate virtual warehouses per workload**. Virtual warehouses isolate compute resources while accessing the same stored data.
- **S2-Q142** (M) A query reads many partitions but returns very few rows. Which optimization topic should the team investigate first? -> **Pruning and clustering effectiveness**. High scan relative to output points toward data-pruning inefficiency.
- **S2-Q147** (M) A business user should see a summarized chart, but not receive unrestricted access to underlying tables. Which principle should the solution preserve? -> **Role-based access control**. RBAC should control access to the underlying Snowflake objects.
- **S2-Q156** (M) A dashboard query result is stale because source data changed. Why might the result cache no longer be reusable? -> **Underlying data changed**. Cache reuse depends on

eligibility, including unchanged underlying data.

- **S2-Q157** (M) A BI team repeats the same expensive query on stable data all day. Before resizing the warehouse, which feature may already reduce repeated compute? -> **Result cache**. Result cache may serve repeated eligible queries without re-executing them.
- **S2-Q167** (H) A dashboard workload has 500 concurrent short queries. Scaling a single warehouse up improves individual-query speed but queuing remains. Which strategy better addresses the root issue? -> **Scale out with a multi-cluster warehouse**. Concurrency pressure is a scale-out problem that multi-cluster warehouses are designed to address.
- **S2-Q168** (H) ETL and BI teams use the same warehouse. During nightly processing, dashboard latency becomes unacceptable. What architecture change best isolates the workloads? -> **Use separate virtual warehouses over the same data**. Separate warehouses isolate compute contention without duplicating stored data.
- **S2-Q169** (H) A query scans many micro-partitions even though predicates are highly selective. Which pair of areas should be reviewed before simply scaling compute? -> **Pruning effectiveness and clustering strategy**. Poor pruning is a data-organization issue; clustering may improve it for repeated filter patterns.
- **S2-Q170** (H) A team adds a clustering key to every large table. Costs rise and some queries show little improvement. What is the best interpretation? -> **Clustering should be targeted to tables and columns with demonstrated pruning benefit**. Clustering is useful when it materially improves pruning for common access patterns, not as a blanket rule.
- **S2-Q171** (H) A repeated report query was fast yesterday due to cache reuse but is slow today after source data was refreshed. What is the most plausible reason? -> **The cached result is no longer eligible because underlying data changed**. Result cache reuse depends on unchanged relevant data and other eligibility conditions.
- **S2-Q196** (H) A company wants low-cost idle periods and rapid restart for bursty analytics. Which warehouse configuration pairing best fits? -> **AUTO_SUSPEND with AUTO_RESUME**. AUTO_SUSPEND saves idle credits and AUTO_RESUME restarts compute when needed.
- **S2-Q197** (H) A company uses separate warehouses for ETL and BI but both still read the same data. Which Snowflake architecture principle makes this possible? -> **Compute and storage are decoupled**. Snowflake separates compute from shared storage.

T3. Supported Clouds; Joins

Exam: 2 MCQ(s) x 1 mark(s), Easy row. Bank: ***23 questions*** *(Set 1: 23, Set 2: 0; Easy 9, Medium 9, Hard 5).*

Memory hooks

- **Clouds:** Snowflake runs on **AWS, Azure and GCP**; the **Region** ties the account to a cloud

geography.

- **Continuity:** Account Replication + **Secondary Database** (replicated copy for failover).
- **Joins:** INNER = matches on both sides; LEFT = keep every left row; CROSS = every combination (Cartesian); SELF = same table with two aliases.
- **Filter before joining** to shrink the join input.

Answer sheet (9 distinct answers, most-asked first)

Answer	Asked	Trigger (what the question is really asking)
INNER JOIN	3	return rows where the join condition matches on both sides / keep only matching records from two related datasets
Account Replication	3	support resilience and continuity across regions or clouds / replicate supported account objects across regions or clouds
AWS, Azure, and GCP	3	choose among the supported cloud platforms while retaining a consistent Snowflake experience / provide Snowflake deployment choices across its supported public clouds
Snowflake Region	3	align account deployment with a specific cloud geography / determine the cloud region where account data is hosted
Secondary Database	3	maintain a replicated database copy for failover scenarios / provide a replicated database target in another region
Filter Before Joining	2	reduce rows before an expensive join when predicates can be applied early / lower the amount of data participating in a large join
LEFT JOIN	2	retain all primary entities even when optional related data is missing / return every row from the left table plus matching rows from the right
CROSS JOIN	2	produce the Cartesian product of two input tables / generate every combination of rows between two small sets
SELF JOIN	2	join a table to itself using aliases / compare rows within the same table, such as employee-manager relationships

All 23 questions with answers

- **S1-Q15** (E) Which Snowflake concept is primarily used to reduce rows before an expensive join when predicates can be applied early? -> **Filter Before Joining**. Early filtering can reduce join input and improve efficiency.

- **S1-Q29** (E) Which Snowflake concept is primarily used to return rows where the join condition matches on both sides? -> **INNER JOIN**. INNER JOIN returns matching rows from both inputs.
- **S1-Q30** (M) A travel team needs to retain all primary entities even when optional related data is missing. Which Snowflake option best meets this requirement? -> **LEFT JOIN**. LEFT JOIN preserves all left-side rows.
- **S1-Q38** (E) Which option best matches this Snowflake responsibility: produce the Cartesian product of two input tables? -> **CROSS JOIN**. CROSS JOIN returns all possible row combinations.
- **S1-Q62** (M) An architect for a insurance system has one key requirement: support resilience and continuity across regions or clouds. Which option best satisfies it? -> **Account Replication**. Account replication supports cross-region/cross-cloud continuity for supported objects.
- **S1-Q90** (M) A manufacturing team needs to lower the amount of data participating in a large join. Which Snowflake option best meets this requirement? -> **Filter Before Joining**. Early filtering can reduce join input and improve efficiency.
- **S1-Q110** (H) A media team is troubleshooting a design. The required outcome is to choose among the supported cloud platforms while retaining a consistent Snowflake experience, but the current approach does not provide that behavior. Which option should replace the current approach? -> **AWS, Azure, and GCP**. Snowflake is available on AWS, Microsoft Azure, and Google Cloud Platform.
- **S1-Q112** (M) A digital payments team needs to align account deployment with a specific cloud geography. Which Snowflake option best meets this requirement? -> **Snowflake Region**. A Snowflake region maps the account to a specific cloud-provider region.
- **S1-Q115** (E) A trainee needs to identify the Snowflake feature that can maintain a replicated database copy for failover scenarios. Which choice is correct? -> **Secondary Database**. A secondary database is used as a replicated copy of a primary database.
- **S1-Q127** (M) In a marketing workload, engineers want to provide a replicated database target in another region. What should they use? -> **Secondary Database**. A secondary database is used as a replicated copy of a primary database.
- **S1-Q128** (H) A digital payments platform must keep only matching records from two related datasets. The team wants the most direct Snowflake mechanism rather than a workaround. Which choice is best? -> **INNER JOIN**. INNER JOIN returns matching rows from both inputs.
- **S1-Q145** (E) A trainee needs to identify the Snowflake feature that can join a table to itself using aliases. Which choice is correct? -> **SELF JOIN**. A self join uses multiple aliases of the same table.
- **S1-Q151** (E) A trainee needs to identify the Snowflake feature that can return every row from the left table plus matching rows from the right. Which choice is correct? -> **LEFT JOIN**. LEFT JOIN preserves all left-side rows.
- **S1-Q161** (E) Which Snowflake concept is primarily used to determine the cloud region where account data is hosted? -> **Snowflake Region**. A Snowflake region maps the account to a

specific cloud-provider region.

- **S1-Q162** (M) In a education workload, engineers want to generate every combination of rows between two small sets. What should they use? -> **CROSS JOIN**. CROSS JOIN returns all possible row combinations.
- **S1-Q163** (M) A financial services team needs to choose among the supported cloud platforms while retaining a consistent Snowflake experience. Which Snowflake option best meets this requirement? -> **AWS, Azure, and GCP**. Snowflake is available on AWS, Microsoft Azure, and Google Cloud Platform.
- **S1-Q166** (H) A supply chain team is troubleshooting a design. The required outcome is to align account deployment with a specific cloud geography, but the current approach does not provide that behavior. Which option should replace the current approach? -> **Snowflake Region**. A Snowflake region maps the account to a specific cloud-provider region.
- **S1-Q179** (E) A trainee needs to identify the Snowflake feature that can replicate supported account objects across regions or clouds. Which choice is correct? -> **Account Replication**. Account replication supports cross-region/cross-cloud continuity for supported objects.
- **S1-Q184** (M) A e-commerce team needs to keep only matching records from two related datasets. Which Snowflake option best meets this requirement? -> **INNER JOIN**. INNER JOIN returns matching rows from both inputs.
- **S1-Q186** (H) A digital payments engineer is asked to meet this constraint: support resilience and continuity across regions or clouds. Which option addresses the constraint with the least unnecessary complexity? -> **Account Replication**. Account replication supports cross-region/cross-cloud continuity for supported objects.
- **S1-Q190** (E) A trainee needs to identify the Snowflake feature that can provide Snowflake deployment choices across its supported public clouds. Which choice is correct? -> **AWS, Azure, and GCP**. Snowflake is available on AWS, Microsoft Azure, and Google Cloud Platform.
- **S1-Q192** (H) A logistics team is troubleshooting a design. The required outcome is to provide a replicated database target in another region, but the current approach does not provide that behavior. Which option should replace the current approach? -> **Secondary Database**. A secondary database is used as a replicated copy of a primary database.
- **S1-Q195** (M) An architect for a education system has one key requirement: compare rows within the same table, such as employee-manager relationships. Which option best satisfies it? -> **SELF JOIN**. A self join uses multiple aliases of the same table.

T4. Snowflake SQL: DDL, DML

*Exam: 4 MCQ(s) x 1 mark(s), Easy row. Bank: **36 questions** (Set 1: 36, Set 2: 0; Easy 14, Medium 14, Hard 8).*

Memory hooks

- **DDL** (structure): CREATE TABLE, CREATE SCHEMA, `DROP TABLE IF EXISTS` (no error if missing), **CREATE OR REPLACE TABLE** (replaces, old data can be lost).
- **TRUNCATE keeps the table, REPLACE removes it.** TRUNCATE = delete all rows but keep the definition; DELETE FROM = only rows that match a condition.
- **DML:** INSERT INTO (add rows), UPDATE (change rows), DELETE, **MERGE** (matched/unmatched logic, upserts), COMMIT (makes the transaction durable).
- **Functions:** LENGTH (characters), COUNT (rows/values), COALESCE (first non-NULL), DATE_PART('month', col) (extract a date part).

Answer sheet (14 distinct answers, most-asked first)

Answer	Asked	Trigger (what the question is really asking)
CREATE SCHEMA	3	organize related database objects under a new schema / create a schema container inside a database
DELETE FROM	3	delete only records that meet a row condition / remove selected rows while keeping the table object
MERGE	3	synchronize a target table with a source using matched and unmatched conditions / perform conditional insert/update logic for incremental data
TRUNCATE TABLE	3	empty a table efficiently without dropping the object / remove all rows while preserving the table structure
UPDATE	3	modify values in existing rows / change selected columns for rows that match a condition
COMMIT	3	make transactional DML changes durable / permanently finalize changes in an explicit transaction
INSERT INTO	3	append new records without changing existing rows / add new rows to an existing table
CREATE TABLE	3	define a new relation for persistent table data / create a new table object
LENGTH()	2	validate or analyze text length / return the number of characters in a string

Answer	Asked	Trigger (what the question is really asking)
DATE_PART('month', date_col)	2	extract the month component from a date or timestamp / derive a month value for grouping time-based reporting
COUNT()	2	measure record counts in an aggregation / return the number of rows or non-NULL expressions according to its argument
DROP TABLE IF EXISTS	2	drop a table only when it exists without raising a missing-object error / make cleanup scripts idempotent when a table may already be absent
CREATE OR REPLACE TABLE	2	replace a table definition and potentially remove the prior table data / recreate a table object when replacing the existing definition is intentional
COALESCE()	2	provide a fallback value when earlier expressions are NULL / return the first non-NULL expression in a list

All 36 questions with answers

- **S1-Q7** (M) An architect for a e-commerce system has one key requirement: validate or analyze text length. Which option best satisfies it? -> **LENGTH()**. LENGTH returns a string length.
- **S1-Q16** (H) A supply chain platform must organize related database objects under a new schema. The team wants the most direct Snowflake mechanism rather than a workaround. Which choice is best? -> **CREATE SCHEMA**. CREATE SCHEMA creates a schema.
- **S1-Q18** (E) Which option best matches this Snowflake responsibility: extract the month component from a date or timestamp? -> **DATE_PART('month', date_col)**. DATE_PART can extract a requested date/time part such as month.
- **S1-Q26** (M) A media platform is being redesigned so it can measure record counts in an aggregation. Which choice is most appropriate? -> **COUNT()**. COUNT is the standard aggregate for counting rows or values.
- **S1-Q41** (E) Which option best matches this Snowflake responsibility: drop a table only when it exists without raising a missing-object error? -> **DROP TABLE IF EXISTS**. IF EXISTS prevents an error when the target table does not exist.
- **S1-Q44** (M) A manufacturing team needs to delete only records that meet a row condition. Which Snowflake option best meets this requirement? -> **DELETE FROM**. DELETE FROM removes qualifying rows.
- **S1-Q52** (E) Which Snowflake capability directly supports the need to synchronize a target table with a source using matched and unmatched conditions? -> **MERGE**. MERGE supports matched/unmatched DML logic against a target table.

- **S1-Q54** (M) A manufacturing platform is being redesigned so it can perform conditional insert/update logic for incremental data. Which choice is most appropriate? -> **MERGE**. MERGE supports matched/unmatched DML logic against a target table.
- **S1-Q55** (M) A travel team needs to derive a month value for grouping time-based reporting. Which Snowflake option best meets this requirement? -> **DATE_PART('month', date_col)**. DATE_PART can extract a requested date/time part such as month.
- **S1-Q65** (M) A marketing platform is being redesigned so it can empty a table efficiently without dropping the object. Which choice is most appropriate? -> **TRUNCATE TABLE**. TRUNCATE TABLE removes all rows while keeping the table definition.
- **S1-Q70** (H) A travel platform must delete only records that meet a row condition. The team wants the most direct Snowflake mechanism rather than a workaround. Which choice is best? -> **DELETE FROM**. DELETE FROM removes qualifying rows.
- **S1-Q75** (E) For the requirement "remove all rows while preserving the table structure," which Snowflake term is the best fit? -> **TRUNCATE TABLE**. TRUNCATE TABLE removes all rows while keeping the table definition.
- **S1-Q80** (E) For the requirement "modify values in existing rows," which Snowflake term is the best fit? -> **UPDATE**. UPDATE changes values in existing rows.
- **S1-Q81** (H) A media team is troubleshooting a design. The required outcome is to make transactional DML changes durable, but the current approach does not provide that behavior. Which option should replace the current approach? -> **COMMIT**. COMMIT finalizes the current transaction.
- **S1-Q82** (M) The data team at a e-commerce company must make cleanup scripts idempotent when a table may already be absent. Which Snowflake feature or command should they select? -> **DROP TABLE IF EXISTS**. IF EXISTS prevents an error when the target table does not exist.
- **S1-Q89** (E) For the requirement "return the number of characters in a string," which Snowflake term is the best fit? -> **LENGTH()**. LENGTH returns a string length.
- **S1-Q91** (H) A e-commerce team is troubleshooting a design. The required outcome is to append new records without changing existing rows, but the current approach does not provide that behavior. Which option should replace the current approach? -> **INSERT INTO**. INSERT INTO adds rows to a table.
- **S1-Q94** (E) Which option best matches this Snowflake responsibility: replace a table definition and potentially remove the prior table data? -> **CREATE OR REPLACE TABLE**. CREATE OR REPLACE TABLE replaces the existing table object, so prior data can be removed.
- **S1-Q106** (M) An architect for a manufacturing system has one key requirement: provide a fallback value when earlier expressions are NULL. Which option best satisfies it? -> **COALESCE()**. COALESCE returns the first non-NULL argument.
- **S1-Q107** (E) For the requirement "add new rows to an existing table," which Snowflake term is the best fit? -> **INSERT INTO**. INSERT INTO adds rows to a table.

- **S1-Q109** (E) Which Snowflake capability directly supports the need to remove selected rows while keeping the table object? -> **DELETE FROM**. DELETE FROM removes qualifying rows.
- **S1-Q113** (E) Which option best matches this Snowflake responsibility: return the first non-NULL expression in a list? -> **COALESCE()**. COALESCE returns the first non-NULL argument.
- **S1-Q121** (M) In a insurance workload, engineers want to recreate a table object when replacing the existing definition is intentional. What should they use? -> **CREATE OR REPLACE TABLE**. CREATE OR REPLACE TABLE replaces the existing table object, so prior data can be removed.
- **S1-Q123** (H) A e-commerce platform must perform conditional insert/update logic for incremental data. The team wants the most direct Snowflake mechanism rather than a workaround. Which choice is best? -> **MERGE**. MERGE supports matched/unmatched DML logic against a target table.
- **S1-Q126** (E) A trainee needs to identify the Snowflake feature that can return the number of rows or non-NULL expressions according to its argument. Which choice is correct? -> **COUNT()**. COUNT is the standard aggregate for counting rows or values.
- **S1-Q136** (H) During a design review for a marketing solution, the team must ensure it can define a new relation for persistent table data. Which Snowflake option is the strongest architectural fit? -> **CREATE TABLE**. CREATE TABLE creates a table object.
- **S1-Q142** (H) A e-commerce team is troubleshooting a design. The required outcome is to change selected columns for rows that match a condition, but the current approach does not provide that behavior. Which option should replace the current approach? -> **UPDATE**. UPDATE changes values in existing rows.
- **S1-Q144** (M) A travel platform is being redesigned so it can organize related database objects under a new schema. Which choice is most appropriate? -> **CREATE SCHEMA**. CREATE SCHEMA creates a schema.
- **S1-Q152** (H) A education platform must empty a table efficiently without dropping the object. The team wants the most direct Snowflake mechanism rather than a workaround. Which choice is best? -> **TRUNCATE TABLE**. TRUNCATE TABLE removes all rows while keeping the table definition.
- **S1-Q155** (E) A trainee needs to identify the Snowflake feature that can create a new table object. Which choice is correct? -> **CREATE TABLE**. CREATE TABLE creates a table object.
- **S1-Q165** (M) A telecommunications platform is being redesigned so it can define a new relation for persistent table data. Which choice is most appropriate? -> **CREATE TABLE**. CREATE TABLE creates a table object.
- **S1-Q172** (E) A trainee needs to identify the Snowflake feature that can create a schema container inside a database. Which choice is correct? -> **CREATE SCHEMA**. CREATE SCHEMA creates a schema.
- **S1-Q185** (M) In a banking workload, engineers want to append new records without changing existing rows. What should they use? -> **INSERT INTO**. INSERT INTO adds rows to a table.

- **S1-Q189** (E) Which Snowflake capability directly supports the need to permanently finalize changes in an explicit transaction? -> **COMMIT**. COMMIT finalizes the current transaction.
- **S1-Q191** (M) The data team at a e-commerce company must make transactional DML changes durable. Which Snowflake feature or command should they select? -> **COMMIT**. COMMIT finalizes the current transaction.
- **S1-Q200** (M) In a SaaS workload, engineers want to change selected columns for rows that match a condition. What should they use? -> **UPDATE**. UPDATE changes values in existing rows.

T5. Visualizations (Matplotlib/Seaborn); SQL with Snowpark

Exam: 4 MCQ(s) x 1 mark(s), Easy row. Bank: ***27 questions*** *(Set 1: 0, Set 2: 27; Easy 14, Medium 11, Hard 2).*

Memory hooks

- **Snowpark DataFrame verbs:** `select()` = keep columns, `filter()` = keep rows, `groupBy()` = before totals, `union()` = stack rows of same-shaped frames, `join()` = match keys, `withColumnRenamed()` = rename.
- Snowpark = server-side processing **inside Snowflake**: less data movement. Never collect a huge table to the client first; embed existing SQL with Snowpark when you want both.
- **Matplotlib** = static charts; **Seaborn** = built on Matplotlib (statistical plots). **Bar** = compare categories, **scatter** = relationship of two numbers.
- Always **filter/aggregate/clean in Snowflake first**, then plot (duplicates make misleading charts).

Answer sheet (21 distinct answers, most-asked first)

Answer	Asked	Trigger (what the question is really asking)
select()	2	Which Snowpark DataFrame method is used when only chosen columns should remain in the result? / A data engineer has a Snowpark DataFrame with 40 columns but a downstream step needs only customer_id, region, and risk_score. Which operat…
filter()	2	Which Snowpark DataFrame operation is designed to retain rows that meet a condition? / A Snowpark pipeline needs to keep orders with amount > 10000

Answer	Asked	Trigger (what the question is really asking)
		before performing an aggregation. Which operation should be used for the row r…
groupBy()	2	Which Snowpark operation is typically used before calculating totals by category? / A healthcare analyst needs total claims by provider using Snowpark. Which operation should organize rows before SUM is applied?
union()	2	Two compatible Snowpark DataFrames contain the same columns and must be stacked row-wise. Which operation fits? / A retailer must combine monthly Snowpark DataFrames with identical schemas into one annual dataset. Which operation is most appropriate?
Matplotlib	2	Seaborn is built on top of which Python plotting library? / A trainer wants a quick static comparison of revenue across departments from a Pandas result. Which library in the source is appropriate?
Scatter plot	2	Which chart is most useful for examining the relationship between two numeric variables? / A dataset contains ad_spend and conversions for many campaigns. Which plot is most suitable for examining their relationship?
withColumnRenamed()	1	Which Snowpark method is commonly used to change a column name without rebuilding the entire DataFrame?
A server-side data-processing framework	1	Snowpark is most accurately described as what kind of capability?
Less data movement between systems	1	What is a major advantage of keeping Snowpark transformations inside Snowflake?
Creating static visualizations	1	Matplotlib is most directly associated with which task?

Answer	Asked	Trigger (what the question is really asking)
Bar chart	1	Which chart type is generally suitable for comparing values across categories?
Filter and aggregate the data appropriately	1	Before plotting business metrics, which preparation step usually improves clarity?
To reuse SQL logic alongside DataFrame transformations	1	Why might a Snowpark developer embed existing SQL inside a programmatic workflow?
join()	1	Which Snowpark DataFrame operation combines columns from related datasets using matching keys?
Use Snowpark DataFrames and reusable Python logic	1	A banking team must apply Python-based fraud rules to transaction data while keeping the data inside Snowflake. The logic includes multiple…
Seaborn	1	An analyst wants a higher-level statistical plotting interface while still using the Matplotlib ecosystem. Which library fits?
Prepare and validate the dataset	1	A bar chart shows misleading totals because duplicate rows were not removed. What should happen before visualization?
Use SQL together with Snowpark DataFrames	1	An analyst wants to combine an existing SQL query with additional Python transformations in the same Snowpark workflow. Which capability su…
Transform/aggregate in Snowflake, then visualize the prepared result	1	A team wants to keep its transformation logic in Snowflake but expose the result in a visualization library. What is the preferred sequence?
Keep transformations in Snowpark so execution stays in Snowflake	1	A Snowpark job transforms a very large dataset, but developers first collect the entire table to the client before processing. Which redesi…
Filter/aggregate in Snowflake, then use a scatter plot on the prepared result	1	A team wants a visual relationship analysis between two continuous variables after filtering millions of rows. Which workflow is best?

All 27 questions with answers

- **S2-Q9** (E) Which Snowpark DataFrame method is used when only chosen columns should remain in the result? -> **select()**. select() projects the requested columns from a DataFrame.
- **S2-Q10** (E) Which Snowpark DataFrame operation is designed to retain rows that meet a condition? -> **filter()**. filter() (or where()) applies a row-level condition.
- **S2-Q11** (E) Which Snowpark operation is typically used before calculating totals by category? -> **groupBy()**. groupBy() creates grouping keys before aggregate calculations.
- **S2-Q12** (E) Two compatible Snowpark DataFrames contain the same columns and must be stacked row-wise. Which operation fits? -> **union()**. union() combines compatible DataFrames vertically.
- **S2-Q13** (E) Which Snowpark method is commonly used to change a column name without rebuilding the entire DataFrame? -> **withColumnRenamed()**. withColumnRenamed() renames a DataFrame column.
- **S2-Q14** (E) Snowpark is most accurately described as what kind of capability? -> **A server-side data-processing framework**. Snowpark lets developers express transformations programmatically while execution occurs in Snowflake.
- **S2-Q15** (E) What is a major advantage of keeping Snowpark transformations inside Snowflake? -> **Less data movement between systems**. Processing close to the data reduces movement to external runtimes.
- **S2-Q48** (E) Matplotlib is most directly associated with which task? -> **Creating static visualizations**. Matplotlib is a Python visualization library.
- **S2-Q49** (E) Seaborn is built on top of which Python plotting library? -> **Matplotlib**. Seaborn extends Matplotlib with higher-level statistical plotting.
- **S2-Q50** (E) Which chart type is generally suitable for comparing values across categories? -> **Bar chart**. Bar charts are commonly used to compare categorical values.
- **S2-Q51** (E) Which chart is most useful for examining the relationship between two numeric variables? -> **Scatter plot**. Scatter plots visualize relationships between two numerical variables.
- **S2-Q52** (E) Before plotting business metrics, which preparation step usually improves clarity? -> **Filter and aggregate the data appropriately**. Filtering and aggregation prepare data for meaningful visualization.
- **S2-Q76** (E) Why might a Snowpark developer embed existing SQL inside a programmatic workflow? -> **To reuse SQL logic alongside DataFrame transformations**. Snowpark can combine SQL logic with programmatic transformations.
- **S2-Q79** (E) Which Snowpark DataFrame operation combines columns from related datasets using matching keys? -> **join()**. join() combines related DataFrames based on matching conditions or keys.
- **S2-Q81** (M) A banking team must apply Python-based fraud rules to transaction data while keeping the data inside Snowflake. The logic includes multiple transformations and reusable

functions. Which approach best fits? -> **Use Snowpark DataFrames and reusable Python logic**. Snowpark supports programmatic transformations that execute close to data inside Snowflake.

- **S2-Q82** (M) A data engineer has a Snowpark DataFrame with 40 columns but a downstream step needs only customer_id, region, and risk_score. Which operation should be applied first? -> **select()**. select() performs column projection and keeps only the requested fields.
- **S2-Q83** (M) A Snowpark pipeline needs to keep orders with amount > 10000 before performing an aggregation. Which operation should be used for the row restriction? -> **filter()**. filter() applies a condition to rows before later transformations.
- **S2-Q84** (M) A retailer must combine monthly Snowpark DataFrames with identical schemas into one annual dataset. Which operation is most appropriate? -> **union()**. union() stacks compatible DataFrames row-wise.
- **S2-Q85** (M) A healthcare analyst needs total claims by provider using Snowpark. Which operation should organize rows before SUM is applied? -> **groupBy()**. groupBy() defines the grouping keys used by aggregate functions.
- **S2-Q102** (M) A trainer wants a quick static comparison of revenue across departments from a Pandas result. Which library in the source is appropriate? -> **Matplotlib**. Matplotlib is used for static Python visualizations.
- **S2-Q103** (M) An analyst wants a higher-level statistical plotting interface while still using the Matplotlib ecosystem. Which library fits? -> **Seaborn**. Seaborn provides high-level statistical visualization on top of Matplotlib.
- **S2-Q104** (M) A dataset contains ad_spend and conversions for many campaigns. Which plot is most suitable for examining their relationship? -> **Scatter plot**. Scatter plots are appropriate for relationships between two numeric variables.
- **S2-Q105** (M) A bar chart shows misleading totals because duplicate rows were not removed. What should happen before visualization? -> **Prepare and validate the dataset**. Accurate visualization depends on correct data preparation and aggregation.
- **S2-Q145** (M) An analyst wants to combine an existing SQL query with additional Python transformations in the same Snowpark workflow. Which capability supports this? -> **Use SQL together with Snowpark DataFrames**. The source highlights combining existing SQL logic with programmatic Snowpark transformations.
- **S2-Q146** (M) A team wants to keep its transformation logic in Snowflake but expose the result in a visualization library. What is the preferred sequence? -> **Transform/aggregate in Snowflake, then visualize the prepared result**. Processing near the data reduces movement and produces cleaner visualization inputs.
- **S2-Q161** (H) A Snowpark job transforms a very large dataset, but developers first collect the entire table to the client before processing. Which redesign best aligns with the source? -> **Keep transformations in Snowpark so execution stays in Snowflake**. Snowpark is intended to process data inside Snowflake and reduce unnecessary movement to the client.

- **S2-Q191** (H) A team wants a visual relationship analysis between two continuous variables after filtering millions of rows. Which workflow is best? -> **Filter/aggregate in Snowflake, then use a scatter plot on the prepared result**. Process data close to the source, then use an appropriate visualization such as a scatter plot.

T6. Snowflake Streamlit

*Exam: 2 MCQ(s) x 1 mark(s), Easy row. Bank: **17 questions** (Set 1: 0, Set 2: 17; Easy 8, Medium 7, Hard 2).*

Memory hooks

- **Streamlit in Snowflake** = interactive data apps that **run inside Snowflake**; access is controlled by **Snowflake roles & privileges** (never hard-code authorization in Python).
- Functions: `st.dataframe()` = table, `st.markdown()` = formatted text, `st.slider()` = numeric range, selectbox = pick one value.
- On every interaction the script **reruns top to bottom**. Aggregate before charting to avoid expensive repeated queries.
- Pairing: **Snowsight dashboards for analytics, Streamlit for interactive apps.**

Answer sheet (16 distinct answers, most-asked first)

Answer	Asked	Trigger (what the question is really asking)
st.dataframe()	2	Which Streamlit function is well suited for rendering a table-like DataFrame? / A Streamlit page should display a filtered DataFrame with sortable columns. Which function is the natural choice?
Interactive data applications	1	What is Streamlit in Snowflake mainly used to build?
Within the Snowflake platform	1	Where does a Streamlit app in Snowflake execute?
st.markdown()	1	Which Streamlit function is intended for formatted Markdown text?
From top to bottom	1	How does a basic Streamlit script typically execute after an interaction triggers a rerun?

Answer	Asked	Trigger (what the question is really asking)
Snowflake roles and privileges	1	Access to Streamlit apps in Snowflake is primarily governed through:
A selection widget such as selectbox	1	Which Streamlit component type is used when a user must choose one value from a list?
st.slider()	1	Which Streamlit widget is designed for choosing a numeric value along a range?
Streamlit in Snowflake	1	A service desk wants a simple app where users choose a region and see summarized Snowflake data interactively. Which capability is most sui…
The script is reevaluated from top to bottom	1	A Streamlit app reruns after a user changes a selection. What execution behavior should a developer expect?
Roles and privileges	1	A governance team must restrict access to a Streamlit app by business role. Which Snowflake mechanism should be used?
Snowsight dashboards for analytics; Streamlit for app interaction	1	A company wants dashboards in Snowsight for analysts and an interactive guided application for business users. Which pairing is most approp…
Streamlit	1	A Snowflake app needs a user-selectable filter, a chart, and a data table. Which source-supported framework best matches?
Existing Snowflake roles and privileges	1	A Streamlit app needs governed data access and should not bypass Snowflake security. What should control permissions?
It bypasses centralized Snowflake RBAC governance	1	A Snowflake app provides a polished UI but hard-codes authorization in Python. Why is this design weak?
Prepare and aggregate only the data needed before visualization	1	A Streamlit app triggers expensive queries on every widget interaction. Which improvement is most aligned with the source concepts?

All 17 questions with answers

- **S2-Q42** (E) What is Streamlit in Snowflake mainly used to build? -> **Interactive data applications**. Streamlit is used for interactive Python-based data apps.
- **S2-Q43** (E) Where does a Streamlit app in Snowflake execute? -> **Within the Snowflake platform**. Streamlit in Snowflake runs within the Snowflake environment.
- **S2-Q44** (E) Which Streamlit function is well suited for rendering a table-like DataFrame? -> **st.dataframe()**. st.dataframe() displays tabular data interactively.
- **S2-Q45** (E) Which Streamlit function is intended for formatted Markdown text? -> **st.markdown()**. st.markdown() renders Markdown-formatted text.
- **S2-Q46** (E) How does a basic Streamlit script typically execute after an interaction triggers a rerun? -> **From top to bottom**. The Streamlit execution model commonly reruns the script from top to bottom.
- **S2-Q47** (E) Access to Streamlit apps in Snowflake is primarily governed through: -> **Snowflake roles and privileges**. Snowflake RBAC controls access to platform objects including applications.
- **S2-Q77** (E) Which Streamlit component type is used when a user must choose one value from a list? -> **A selection widget such as selectbox**. Streamlit selection widgets support interactive user choices.
- **S2-Q80** (E) Which Streamlit widget is designed for choosing a numeric value along a range? -> **st.slider()**. st.slider() provides an interactive numeric range/value selector. MEDIUM QUESTIONS
- **S2-Q98** (M) A service desk wants a simple app where users choose a region and see summarized Snowflake data interactively. Which capability is most suitable? -> **Streamlit in Snowflake**. Streamlit is intended for interactive data applications inside Snowflake.
- **S2-Q99** (M) A Streamlit page should display a filtered DataFrame with sortable columns. Which function is the natural choice? -> **st.dataframe()**. st.dataframe() is designed for tabular DataFrame display.
- **S2-Q100** (M) A Streamlit app reruns after a user changes a selection. What execution behavior should a developer expect? -> **The script is reevaluated from top to bottom**. The standard Streamlit model reruns the script from top to bottom on interaction.
- **S2-Q101** (M) A governance team must restrict access to a Streamlit app by business role. Which Snowflake mechanism should be used? -> **Roles and privileges**. Access to Snowflake objects is governed with RBAC.
- **S2-Q143** (M) A company wants dashboards in Snowsight for analysts and an interactive guided application for business users. Which pairing is most appropriate? -> **Snowsight dashboards for analytics; Streamlit for app interaction**. Snowsight suits dashboards/SQL analysis, while Streamlit suits interactive applications.
- **S2-Q144** (M) A Snowflake app needs a user-selectable filter, a chart, and a data table. Which source-supported framework best matches? -> **Streamlit**. Streamlit provides interactive

widgets and data display components.

- **S2-Q155** (M) A Streamlit app needs governed data access and should not bypass Snowflake security. What should control permissions? -> **Existing Snowflake roles and privileges**. Streamlit apps in Snowflake should rely on platform RBAC.
- **S2-Q189** (H) A Snowflake app provides a polished UI but hard-codes authorization in Python. Why is this design weak? -> **It bypasses centralized Snowflake RBAC governance**. Platform roles and privileges should remain the primary authorization mechanism.
- **S2-Q190** (H) A Streamlit app triggers expensive queries on every widget interaction. Which improvement is most aligned with the source concepts? -> **Prepare and aggregate only the data needed before visualization**. Efficient data preparation reduces unnecessary processing before visualization.

T7. Snowflake Horizon

*Exam: 2 MCQ(s) x 2 mark(s), Medium row. Bank: **8 questions** (Set 1: 0, Set 2: 8; Easy 2, Medium 4, Hard 2).*

Memory hooks

- **Horizon = governance & security visibility**: who accesses what, policy enforcement, compliance. It is **not** a query-speed feature.
- Pairing to remember: **Horizon for governance, Query Profile for performance.** RBAC is the access-control foundation; Horizon adds visibility on top.

Answer sheet (6 distinct answers, most-asked first)

Answer	Asked	Trigger (what the question is really asking)
Snowflake Horizon	3	A legal team wants organization-wide visibility into governance, data access, and compliance signals inside Snowflake. Which capability is… / An auditor wants to understand how sensitive data is accessed and governed across Snowflake. Which capability from the source is designed f…
Governance and security visibility	1	Snowflake Horizon is most closely associated with which business need?
Better visibility into data access and policy enforcement	1	Which outcome is aligned with Snowflake Horizon?

Answer	Asked	Trigger (what the question is really asking)
Horizon	1	A governance analyst wants to know whether a platform feature is intended for security visibility rather than query acceleration. Which one…
Horizon for governance; Query Profile for query performance	1	An auditor asks for a platform-wide view of data governance and policy visibility, while a performance engineer asks why one query is slow.…
Snowflake Horizon governance visibility	1	A governance lead sees strong RBAC but limited understanding of how data is being accessed across the platform. Which additional capability…

All 8 questions with answers

- **S2-Q53** (E) Snowflake Horizon is most closely associated with which business need? -> **Governance and security visibility**. Horizon focuses on governance, security, compliance, and visibility.
- **S2-Q54** (E) Which outcome is aligned with Snowflake Horizon? -> **Better visibility into data access and policy enforcement**. Horizon uses Snowflake metadata and telemetry to support governance visibility.
- **S2-Q106** (M) A legal team wants organization-wide visibility into governance, data access, and compliance signals inside Snowflake. Which capability is most aligned? -> **Snowflake Horizon**. Horizon focuses on governance, security, and compliance visibility.
- **S2-Q107** (M) An auditor wants to understand how sensitive data is accessed and governed across Snowflake. Which capability from the source is designed for this broader governance need? -> **Snowflake Horizon**. Horizon integrates governance visibility using Snowflake metadata and telemetry.
- **S2-Q140** (M) A compliance lead wants to see access and policy visibility rather than simply query runtime. Which platform capability is better suited? -> **Snowflake Horizon**. Horizon focuses on governance and compliance visibility rather than query tuning alone.
- **S2-Q158** (M) A governance analyst wants to know whether a platform feature is intended for security visibility rather than query acceleration. Which one is it? -> **Horizon**. Horizon is focused on governance/security/compliance rather than pure query acceleration.
- **S2-Q188** (H) An auditor asks for a platform-wide view of data governance and policy visibility, while a performance engineer asks why one query is slow. Which pairing is correct? -> **Horizon for governance; Query Profile for query performance**. Horizon addresses governance visibility, whereas Query Profile addresses query execution behavior.

- **S2-Q199** (H) A governance lead sees strong RBAC but limited understanding of how data is being accessed across the platform. Which additional capability should be emphasized? -> **Snowflake Horizon governance visibility**. Horizon complements access control with governance and usage visibility.

T8. Monitoring Usage; Dynamic Tables

*Exam: 3 MCQ(s) x 2 mark(s), Medium row. Bank: **29 questions** (Set 1: 8, Set 2: 21; Easy 4, Medium 19, Hard 6).*

Memory hooks

- **Dynamic table** = a derived table kept fresh automatically as upstream data changes (little orchestration). **Target lag / refresh policy** controls how fresh; fresher = **more compute/credits**.
- Chains need correct **dependencies** (refresh upstream to downstream). If an upstream table fails or lags, downstream is **delayed or blocked**.
- **Dynamic Tables vs Streams+Tasks:** declarative maintained dataset = Dynamic Table; explicit change capture + custom procedural logic = Streams+Tasks.
- Monitoring (medium): longest queries = Query History; warehouse cost over time = `WAREHOUSE_METERING_HISTORY`; slow/expensive query = **Query Profile**; high bytes scanned but few returned = **poor pruning**. Check Profile & scan metrics **before** resizing.

Answer sheet (27 distinct answers, most-asked first)

Answer	Asked	Trigger (what the question is really asking)
Query Profile	2	diagnose slow operations inside an executed query / A team wants to understand why one SQL statement is expensive relative to another. Which feature should they inspect?
Dynamic Tables	2	Which Snowflake feature maintains derived data automatically as upstream data changes? / An architect must choose between Dynamic Tables and Streams+Tasks. The requirement is a maintained derived dataset with minimal custom orch…

Answer	Asked	Trigger (what the question is really asking)
Resource Monitor	1	control unexpected compute spending across warehouses
Partitions Scanned	1	judge whether filters are pruning storage effectively
QUERY_HISTORY	1	review who ran queries and how they performed
History tab	1	inspect recent query executions from the UI
Execution Graph	1	locate the operator responsible for most query work
Warehouse Usage History	1	analyze warehouse compute consumption over time
Bytes Scanned vs. Bytes Returned	1	identify scans that read much more data than the result requires
When derived data is refreshed	1	What does a dynamic table refresh policy primarily control?
They help preserve the correct refresh order	1	Why are dependencies important in a chain of dynamic tables?
Compute	1	Which resource is consumed when dynamic tables are refreshed?
Snowsight Query History	1	An analyst wants to find yesterday's top five longest-running queries in the web UI. Which Snowflake capability should they use first?
Snowsight monitoring and dashboards	1	A platform owner wants a visual summary of warehouse usage and query behavior without building an external monitoring app. Which interface…
WAREHOUSE_METERING_HISTORY	1	A finance team sees compute cost rising and wants historical warehouse-level usage.

Answer	Asked	Trigger (what the question is really asking)
		Which source is most appropriate?
Dynamic table	1	A platform team wants to reduce manual orchestration for a derived sales summary that should stay current as source tables change. Which fe…
To refresh the chain in a valid upstream-to-downstream order	1	A dynamic table chain contains bronze_summary -> silver_summary -> gold_summary. Why must dependencies be defined correctly?
Refresh policy / target freshness	1	A dynamic table is refreshing more often than the business needs and consuming extra credits. Which design area should be reviewed?
Its refresh may be delayed or blocked	1	An upstream dynamic table fails to refresh. What is the expected effect on a dependent downstream dynamic table?
Query duration and credits consumed	1	A team wants to compare whether doubling warehouse size is worth the performance improvement. Which pair of metrics should be considered to…
Dynamic Tables or Streams+Tasks	1	A team wants to build an incremental data mart without maintaining a custom scheduler and change-detection script. Which built-in alternati…
Refresh policy / target lag	1	A source table receives updates throughout the day, but a derived table only needs freshness within a defined window. Which Dynamic Table d…
Correlate warehouse metering with Query History / Query Profile	1	A platform team sees a cost spike but query volume is unchanged. Which analysis path best separates whether the issue is warehouse configur…

Answer		Asked Trigger (what the question is really asking)
Upstream refresh health and dependency sequencing	1	A dynamic table chain is repeatedly missing its freshness objective because an upstream object is delayed. Which root cause should be inves…
Fresher derived data requires more frequent compute work	1	A team configures a very aggressive Dynamic Table freshness target and sees credit consumption rise sharply. What trade-off is being expose…
Use Query Profile and scan/pruning metrics before resizing compute	1	A business unit repeatedly scales warehouses larger to solve every performance issue. Which diagnostic approach is more disciplined?
Inefficient pruning or overly broad scans	1	A company sees a low amount of returned data but very high scanned bytes. What does this most strongly suggest?

All 29 questions with answers

- **S1-Q33** (M) A financial services team needs to control unexpected compute spending across warehouses. Which Snowflake option best meets this requirement? -> **Resource Monitor**. Resource monitors govern credit consumption and can trigger notifications or suspension actions.
- **S1-Q36** (M) The data team at a telecommunications company must judge whether filters are pruning storage effectively. Which Snowflake feature or command should they select? -> **Partitions Scanned**. Partitions Scanned indicates the micro-partitions read by a query.
- **S1-Q104** (M) A insurance team needs to review who ran queries and how they performed. Which Snowflake option best meets this requirement? -> **QUERY_HISTORY**. QUERY_HISTORY exposes query execution history metadata.
- **S1-Q111** (M) A financial services platform is being redesigned so it can inspect recent query executions from the UI. Which choice is most appropriate? -> **History tab**. The History area provides access to completed query activity.
- **S1-Q116** (M) A travel platform is being redesigned so it can locate the operator responsible for most query work. Which choice is most appropriate? -> **Execution Graph**. The execution graph shows operators and their relationships in a query plan.
- **S1-Q140** (M) In a financial services workload, engineers want to analyze warehouse compute consumption over time. What should they use? -> **Warehouse Usage History**. Warehouse

usage history provides historical warehouse activity and credit information.

- **S1-Q143** (M) An architect for a digital payments system has one key requirement: diagnose slow operations inside an executed query. Which option best satisfies it? -> **Query Profile**. Query Profile breaks down query execution and performance.
- **S1-Q154** (M) In a travel workload, engineers want to identify scans that read much more data than the result requires. What should they use? -> **Bytes Scanned vs. Bytes Returned**. Comparing bytes scanned with bytes returned can reveal inefficient scanning.
- **S2-Q31** (E) Which Snowflake feature maintains derived data automatically as upstream data changes? -> **Dynamic Tables**. Dynamic Tables automatically maintain derived datasets according to refresh rules.
- **S2-Q32** (E) What does a dynamic table refresh policy primarily control? -> **When derived data is refreshed**. Refresh policy governs update timing/behavior for dynamic tables.
- **S2-Q33** (E) Why are dependencies important in a chain of dynamic tables? -> **They help preserve the correct refresh order**. Dependencies help Snowflake refresh upstream and downstream objects in a consistent order.
- **S2-Q34** (E) Which resource is consumed when dynamic tables are refreshed? -> **Compute**. Refresh operations require compute resources.
- **S2-Q91** (M) An analyst wants to find yesterday's top five longest-running queries in the web UI. Which Snowflake capability should they use first? -> **Snowsight Query History**. Query History is the direct place to inspect query duration and execution details.
- **S2-Q92** (M) A platform owner wants a visual summary of warehouse usage and query behavior without building an external monitoring app. Which interface is most appropriate? -> **Snowsight monitoring and dashboards**. Snowsight provides integrated monitoring views and dashboards for usage and performance.
- **S2-Q93** (M) A finance team sees compute cost rising and wants historical warehouse-level usage. Which source is most appropriate? -> **WAREHOUSE_METERING_HISTORY**. WAREHOUSE_METERING_HISTORY exposes warehouse credit-consumption history.
- **S2-Q108** (M) A platform team wants to reduce manual orchestration for a derived sales summary that should stay current as source tables change. Which feature fits best? -> **Dynamic table**. Dynamic tables automatically maintain derived datasets based on upstream changes.
- **S2-Q109** (M) A dynamic table chain contains bronze_summary -> silver_summary -> gold_summary. Why must dependencies be defined correctly? -> **To refresh the chain in a valid upstream-to-downstream order**. Dependencies control refresh sequencing between related dynamic tables.
- **S2-Q110** (M) A dynamic table is refreshing more often than the business needs and consuming extra credits. Which design area should be reviewed? -> **Refresh policy / target freshness**. Refresh behavior controls how frequently compute is used to update derived data.

- **S2-Q111** (M) An upstream dynamic table fails to refresh. What is the expected effect on a dependent downstream dynamic table? -> **Its refresh may be delayed or blocked**. Downstream refreshes depend on successful upstream data availability.
- **S2-Q139** (M) A team wants to compare whether doubling warehouse size is worth the performance improvement. Which pair of metrics should be considered together? -> **Query duration and credits consumed**. Performance/cost tuning requires considering runtime and credit consumption together.
- **S2-Q141** (M) A team wants to understand why one SQL statement is expensive relative to another. Which feature should they inspect? -> **Query Profile**. Query Profile exposes execution behavior for performance analysis.
- **S2-Q159** (M) A team wants to build an incremental data mart without maintaining a custom scheduler and change-detection script. Which built-in alternatives are most relevant? -> **Dynamic Tables or Streams+Tasks**. Dynamic Tables automate refresh, while Streams+Tasks provide an incremental pipeline pattern.
- **S2-Q160** (M) A source table receives updates throughout the day, but a derived table only needs freshness within a defined window. Which Dynamic Table design control matters? -> **Refresh policy / target lag**. Freshness policy determines how current the maintained derived table should be. HARD QUESTIONS
- **S2-Q166** (H) A platform team sees a cost spike but query volume is unchanged. Which analysis path best separates whether the issue is warehouse configuration or query inefficiency? -> **Correlate warehouse metering with Query History / Query Profile**. Metering reveals compute consumption while query history/profile explains execution behavior.
- **S2-Q179** (H) A dynamic table chain is repeatedly missing its freshness objective because an upstream object is delayed. Which root cause should be investigated first? -> **Upstream refresh health and dependency sequencing**. Dependent dynamic tables rely on successful upstream refreshes and correct dependency order.
- **S2-Q180** (H) A team configures a very aggressive Dynamic Table freshness target and sees credit consumption rise sharply. What trade-off is being exposed? -> **Fresher derived data requires more frequent compute work**. More frequent refreshes generally require more compute.
- **S2-Q181** (H) An architect must choose between Dynamic Tables and Streams+Tasks. The requirement is a maintained derived dataset with minimal custom orchestration. Which is the stronger fit? -> **Dynamic Tables**. Dynamic Tables directly maintain derived data based on upstream changes and refresh policy.
- **S2-Q194** (H) A business unit repeatedly scales warehouses larger to solve every performance issue. Which diagnostic approach is more disciplined? -> **Use Query Profile and scan/pruning metrics before resizing compute**. Query diagnostics should identify whether the bottleneck is scan, join, or compute before resizing.

- **S2-Q195** (H) A company sees a low amount of returned data but very high scanned bytes. What does this most strongly suggest? -> **Inefficient pruning or overly broad scans**. Large scans for small results often indicate pruning or filtering inefficiency.

T9. Snowflake Streams; Tasks; Iceberg Tables

*Exam: 1 MCQ(s) x 2 mark(s), Medium row. Bank: **19 questions** (Set 1: 6, Set 2: 13; Easy 8, Medium 8, Hard 3).*

Memory hooks

- **Stream = the spy** (records inserts/updates/deletes so you process only changes). **Task = the timer** (runs SQL/procedures on a schedule, `CRON`, in dependency order).
- **Stream + Task** = standard incremental pipeline (e.g. every 15 min, only when changes exist).
- **Iceberg tables** = Snowflake working with an **open table format** (lakehouse).

Answer sheet (13 distinct answers, most-asked first)

Answer	Asked	Trigger (what the question is really asking)
Tasks	3	schedule or trigger SQL/procedural work in Snowflake / run recurring transformation work without manual execution
Streams	3	process only changed rows instead of rescanning an entire source table / track row-level DML changes for incremental downstream processing
CRON	2	express a recurring calendar schedule for a Snowflake task / run a task at defined calendar times
Iceberg tables	2	A lakehouse-oriented team wants Snowflake to work with an open table format. Which source-supported feature is relevant? / A team wants Snowflake to participate in a lakehouse architecture using an open table format. Which capability from the source is most rele…
Streams and Tasks	1	Which pairing is commonly used to build an incremental pipeline inside Snowflake?
Open table formats	1	Snowflake Iceberg tables are primarily associated with support for:

Answer	Asked	Trigger (what the question is really asking)
Execution order between related tasks	1	What do task dependencies allow a Snowflake pipeline to express?
Stream	1	A customer table changes throughout the day. A downstream job needs only changed rows since the previous run. Which feature should capture…
Stream plus Task	1	A team needs an incremental pipeline that processes stream changes every 15 minutes. Which combination best fits?
Tasks with dependencies	1	A pipeline has three transformations that must run in sequence automatically. Which feature organizes the scheduled steps?
Use Streams with Tasks	1	A logistics pipeline should process a new batch only when changes exist in the source table. Which source-supported pattern is most appropr…
Streams with Tasks	1	An architect must choose between Dynamic Tables and Streams+Tasks. The requirement is explicit change capture plus custom procedural proces…
Stream for change capture plus Tasks for automated processing	1	A source contains frequent row changes, a transformation must run automatically, and downstream consumers need only processed deltas. Which…

All 19 questions with answers

- **S1-Q69** (E) Which Snowflake concept is primarily used to express a recurring calendar schedule for a Snowflake task? -> **CRON**. CRON syntax can be used to define recurring task schedules.
- **S1-Q84** (E) A trainee needs to identify the Snowflake feature that can schedule or trigger SQL/procedural work in Snowflake. Which choice is correct? -> **Tasks**. Tasks automate scheduled or triggered work.
- **S1-Q87** (M) An architect for a marketing system has one key requirement: run a task at defined calendar times. Which option best satisfies it? -> **CRON**. CRON syntax can be used to define recurring task schedules.
- **S1-Q92** (M) In a e-commerce workload, engineers want to process only changed rows instead of rescanning an entire source table. What should they use? -> **Streams**. Streams expose change data for incremental processing.

- **S1-Q101** (E) A trainee needs to identify the Snowflake feature that can track row-level DML changes for incremental downstream processing. Which choice is correct? -> **Streams**. Streams expose change data for incremental processing.
- **S1-Q147** (M) An architect for a education system has one key requirement: run recurring transformation work without manual execution. Which option best satisfies it? -> **Tasks**. Tasks automate scheduled or triggered work.
- **S2-Q35** (E) Which feature records row-level changes such as inserts, updates, and deletes for incremental processing? -> **Streams**. Streams capture change data for downstream processing.
- **S2-Q36** (E) Which feature is designed to schedule SQL or procedural operations? -> **Tasks**. Tasks automate scheduled or dependent execution.
- **S2-Q37** (E) Which pairing is commonly used to build an incremental pipeline inside Snowflake? -> **Streams and Tasks**. Streams identify changed rows and Tasks automate processing.
- **S2-Q61** (E) Snowflake Iceberg tables are primarily associated with support for: -> **Open table formats**. The source identifies Iceberg support as enabling open table formats.
- **S2-Q75** (E) What do task dependencies allow a Snowflake pipeline to express? -> **Execution order between related tasks**. Task dependencies define the order in which related task steps execute.
- **S2-Q112** (M) A customer table changes throughout the day. A downstream job needs only changed rows since the previous run. Which feature should capture those changes? -> **Stream**. Streams track inserts, updates, and deletes for incremental processing.
- **S2-Q113** (M) A team needs an incremental pipeline that processes stream changes every 15 minutes. Which combination best fits? -> **Stream plus Task**. Streams capture changes and Tasks automate processing.
- **S2-Q114** (M) A pipeline has three transformations that must run in sequence automatically. Which feature organizes the scheduled steps? -> **Tasks with dependencies**. Task dependencies support scheduled multi-step orchestration.
- **S2-Q115** (M) A logistics pipeline should process a new batch only when changes exist in the source table. Which source-supported pattern is most appropriate? -> **Use Streams with Tasks**. Streams expose changes and Tasks can automate incremental workflows.
- **S2-Q129** (M) A lakehouse-oriented team wants Snowflake to work with an open table format. Which source-supported feature is relevant? -> **Iceberg tables**. Snowflake Iceberg tables provide support for open table formats.
- **S2-Q182** (H) An architect must choose between Dynamic Tables and Streams+Tasks. The requirement is explicit change capture plus custom procedural processing steps. Which is the stronger fit? -> **Streams with Tasks**. Streams capture changes and Tasks orchestrate custom processing logic.
- **S2-Q187** (H) A team wants Snowflake to participate in a lakehouse architecture using an open table format. Which capability from the source is most relevant? -> **Iceberg tables**. Snowflake

Iceberg tables provide open table format support.

- **S2-Q200** (H) A source contains frequent row changes, a transformation must run automatically, and downstream consumers need only processed deltas. Which architecture is most appropriate? -> **Stream for change capture plus Tasks for automated processing**. Streams+Tasks provide the source-supported pattern for automated incremental processing.

T10. Snowpipe; Snowflake Integration with dbt

*Exam: 2 MCQ(s) x 2 mark(s), Difficult row. Bank: **24 questions** (Set 1: 3, Set 2: 21; Easy 10, Medium 9, Hard 5).*

Memory hooks

- **Snowpipe** = automatic, near-real-time file loading triggered by **cloud storage event notifications**. Nothing loading? **Check the notification configuration first** (manual COPY working proves the data is fine).
- Snowpipe is **not free**: it consumes compute for ingestion even with no warehouse of yours running. Failed files? Check **load history**.
- **dbt** = SQL-based transformation. Dependencies are a **DAG**; use **raw / staging / analytics schemas**; deploy with **CI/CD**; models materialize as **tables or views**.

Answer sheet (18 distinct answers, most-asked first)

Answer	Asked	Trigger (what the question is really asking)
Snowpipe	5	load files soon after they arrive in cloud storage without repeatedly running manual COPY commands / automatically ingest newly arriving staged files with low operational effort
Cloud storage event notifications	2	What usually triggers Snowpipe auto-ingest from cloud object storage? / A Snowpipe pipeline is configured but new files are never detected. Which integration point should be checked first?
Load history	2	Which history information is relevant when checking whether automated file ingestion failed? / An operator needs to investigate which files failed during automated ingestion. Which history information is most relevant?

Answer	Asked	Trigger (what the question is really asking)
Compute used for ingestion	1	Snowpipe consumption is primarily associated with which type of resource?
SQL-based data transformation	1	In the source material, dbt is primarily associated with which activity?
As a DAG	1	How are dbt model dependencies commonly represented?
CI/CD integration	1	Which deployment practice is commonly associated with production dbt workflows?
Separate raw, staging, and analytics layers	1	Which schema design practice is recommended for dbt projects in the source?
Tables or views	1	When a dbt model is materialized in Snowflake, which objects are commonly produced?
Snowpipe consumes compute resources for ingestion	1	A finance manager asks why automated ingestion generates cost even without a user-managed warehouse running. Which explanation fits the sou…
DAG	1	A dbt project contains staging models, intermediate models, and marts. Which dbt concept expresses the dependency order?
Separate raw, staging, and analytics schemas	1	A team is reorganizing a dbt project for maintainability. Which structure is most aligned with the source?
CI/CD pipeline	1	A company wants dbt changes promoted reliably across environments. Which engineering practice should be integrated?
The model materializes as a table or view	1	An analytics engineer defines a dbt model and chooses a persistent relation rather than an ephemeral expression. Which outcome is common?
Auto-ingest event notification configuration	1	A Snowpipe pipeline receives no new rows even though files arrive in the bucket. Manual COPY INTO succeeds. Which area should be checked fi…

Answer	Asked	Trigger (what the question is really asking)
Snowpipe still consumes compute resources for ingestion	1	A team is troubleshooting ingestion cost. They assume Snowpipe is free because no user-managed warehouse is running. What should be correct…
Layered schemas, DAG-managed dependencies, and CI/CD deployment	1	A production dbt project has all models in one schema, manual deployments, and unclear dependencies. Which improvement set best aligns with…
Model dependencies expressed as a DAG	1	A dbt model change breaks a downstream mart. Which built-in design principle makes such relationships visible before runtime?

All 24 questions with answers

- **S1-Q20** (H) A logistics platform must load files soon after they arrive in cloud storage without repeatedly running manual COPY commands. The team wants the most direct Snowflake mechanism rather than a workaround. Which choice is best? -> **Snowpipe**. Snowpipe provides continuous/automated file ingestion.
- **S1-Q86** (M) A retail platform is being redesigned so it can load files soon after they arrive in cloud storage without repeatedly running manual COPY commands. Which choice is most appropriate? -> **Snowpipe**. Snowpipe provides continuous/automated file ingestion.
- **S1-Q173** (E) Which option best matches this Snowflake responsibility: automatically ingest newly arriving staged files with low operational effort? -> **Snowpipe**. Snowpipe provides continuous/automated file ingestion.
- **S2-Q38** (E) Which feature is intended for automated near-real-time file ingestion from cloud storage? -> **Snowpipe**. Snowpipe supports automated ingestion as files arrive.
- **S2-Q39** (E) What usually triggers Snowpipe auto-ingest from cloud object storage? -> **Cloud storage event notifications**. Event notifications indicate that new files are available for ingestion.
- **S2-Q40** (E) Snowpipe consumption is primarily associated with which type of resource? -> **Compute used for ingestion**. Snowpipe charges are primarily tied to compute resources used during ingestion.
- **S2-Q41** (E) Which history information is relevant when checking whether automated file ingestion failed? -> **Load history**. Load history helps diagnose file-loading outcomes.
- **S2-Q57** (E) In the source material, dbt is primarily associated with which activity? -> **SQL-based data transformation**. dbt is used to define SQL transformations and models.
- **S2-Q58** (E) How are dbt model dependencies commonly represented? -> **As a DAG**. dbt models form a directed acyclic graph of dependencies.

- **S2-Q59** (E) Which deployment practice is commonly associated with production dbt workflows? -> **CI/CD integration**. The source links production dbt deployment with CI/CD pipelines.
- **S2-Q60** (E) Which schema design practice is recommended for dbt projects in the source? -> **Separate raw, staging, and analytics layers**. Layered schemas improve organization and maintainability.
- **S2-Q78** (E) When a dbt model is materialized in Snowflake, which objects are commonly produced? -> **Tables or views**. The source identifies tables and views as common dbt model materializations.
- **S2-Q116** (M) A media company receives files at unpredictable times and wants low-latency automatic loading from cloud storage. Which service fits? -> **Snowpipe**. Snowpipe supports automated near-real-time file ingestion.
- **S2-Q117** (M) A Snowpipe pipeline is configured but new files are never detected. Which integration point should be checked first? -> **Cloud storage event notifications**. Auto-ingest relies on event notifications from cloud storage.
- **S2-Q118** (M) An operator needs to investigate which files failed during automated ingestion. Which history information is most relevant? -> **Load history**. Load history records outcomes of file loading operations.
- **S2-Q119** (M) A finance manager asks why automated ingestion generates cost even without a user-managed warehouse running. Which explanation fits the source? -> **Snowpipe consumes compute resources for ingestion**. Snowpipe charging is tied primarily to compute consumed during ingestion.
- **S2-Q125** (M) A dbt project contains staging models, intermediate models, and marts. Which dbt concept expresses the dependency order? -> **DAG**. dbt represents model dependencies as a directed acyclic graph.
- **S2-Q126** (M) A team is reorganizing a dbt project for maintainability. Which structure is most aligned with the source? -> **Separate raw, staging, and analytics schemas**. The source recommends separating raw, staging, and analytics layers.
- **S2-Q127** (M) A company wants dbt changes promoted reliably across environments. Which engineering practice should be integrated? -> **CI/CD pipeline**. The source associates production dbt deployments with CI/CD.
- **S2-Q128** (M) An analytics engineer defines a dbt model and chooses a persistent relation rather than an ephemeral expression. Which outcome is common? -> **The model materializes as a table or view**. dbt models in Snowflake commonly materialize as tables or views.
- **S2-Q183** (H) A Snowpipe pipeline receives no new rows even though files arrive in the bucket. Manual COPY INTO succeeds. Which area should be checked first? -> **Auto-ingest event notification configuration**. If manual loading works but auto-ingest does not, event notification wiring is a likely root cause.
- **S2-Q184** (H) A team is troubleshooting ingestion cost. They assume Snowpipe is free because no user-managed warehouse is running. What should be corrected? -> **Snowpipe still**

consumes compute resources for ingestion. Snowpipe charges are primarily associated with compute used to ingest data.

- **S2-Q185** (H) A production dbt project has all models in one schema, manual deployments, and unclear dependencies. Which improvement set best aligns with the source? -> **Layered schemas, DAG-managed dependencies, and CI/CD deployment**. The source supports layered schemas, DAG dependencies, and CI/CD for production dbt workflows.
- **S2-Q186** (H) A dbt model change breaks a downstream mart. Which built-in design principle makes such relationships visible before runtime? -> **Model dependencies expressed as a DAG**. dbt dependency graphs make upstream/downstream relationships explicit.

T11. Snowflake UDFs

Exam: 2 MCQ(s) x 2 mark(s), Difficult row. Bank: ***19 questions*** *(Set 1: 0, Set 2: 19; Easy 5, Medium 9, Hard 5).*

Memory hooks

- **UDF = a *formula*** returns a value you can use inside a SELECT. **Scalar UDF** = one value per call; table UDF = returns rows. Needs **CREATE FUNCTION** privilege.
- **UDF vs procedure:** loops, branching, several DML steps, orchestration → **stored procedure**. Simple reusable calculation in a SELECT → **UDF**.
- Python/Snowpark UDF slow? Look at **complexity of the logic** and use **vectorized** operations. Deterministic = same input, same output.
- **External function** = call a service outside Snowflake (adds dependency/integration overhead; don't use if a native UDF works).

Answer sheet (17 distinct answers, most-asked first)

Answer	Asked	Trigger (what the question is really asking)
External function	2	Which type of Snowflake function can reach an external service when native execution is not sufficient? / A company wants an external service to be called from Snowflake logic. Which object category from the source is relevant?
UDF	2	A developer needs a reusable expression that returns a computed value but should not perform multi-step orchestration. Which object fits be… / A team is choosing between a UDF and a stored procedure. The requirement is a

Answer	Asked	Trigger (what the question is really asking)
		reusable calculation in a SELECT list. Which should they choo…
User-defined function	1	Which Snowflake object is intended to return a value that can be reused inside expressions?
A scalar UDF returns one value per invocation	1	Which statement best distinguishes a scalar UDF from a table UDF?
CREATE FUNCTION	1	Which privilege is directly associated with defining a new UDF in a schema?
They let reusable programmatic logic execute close to Snowflake data	1	Why are Snowpark UDFs useful for data engineering teams?
Snowpark UDF	1	A team has reusable Python scoring logic that must be called directly from SQL for each input row. Which Snowflake object is most suitable?
The complexity of logic inside the UDF	1	A UDF runs correctly but becomes slow after logic is expanded with many complex operations. Which factor from the source is most directly r…
Use vectorized operations where appropriate	1	A team wants to reduce overhead when processing batches of values in a Python UDF. Which practice is most aligned with the source?
Use a stored procedure instead	1	A SQL UDF is being considered for a workflow that needs loops and branching. What is the better choice?
Deterministic behavior	1	A UDF must produce the same result whenever given the same input values. Which characteristic describes this?
Stored procedure	1	A team is choosing between a UDF and a stored procedure. The requirement is to automate a sequence of SQL steps and control flow. Which sho…

Answer	Asked	Trigger (what the question is really asking)
Simplify logic and use vectorized processing where possible	1	A Python UDF is called millions of times and each call performs expensive row-by-row logic. Which source-supported optimization deserves in…
SQL UDFs are not intended for procedural control-flow workflows	1	A developer proposes a SQL UDF for a process that needs loops, conditional branching, and multiple DML steps. What is the strongest design…
A scalar UDF	1	A team chooses a stored procedure for a simple per-row calculation used in many SELECT statements. Which alternative is more appropriate?
External calls can add dependency and integration overhead	1	An engineer uses an external function for logic that could execute natively as a simple Snowflake UDF. What architectural downside should b…
Scalar UDF for the calculation; stored procedure for orchestration	1	A team needs one reusable function to return a scalar value and another routine to orchestrate multiple statements with conditional branche…

All 19 questions with answers

- **S2-Q5** (E) Which Snowflake object is intended to return a value that can be reused inside expressions? -> **User-defined function**. A UDF encapsulates reusable logic and returns a value that can be referenced in expressions.
- **S2-Q6** (E) Which statement best distinguishes a scalar UDF from a table UDF? -> **A scalar UDF returns one value per invocation**. Scalar UDFs return a single value, whereas table functions can return tabular results.
- **S2-Q7** (E) Which privilege is directly associated with defining a new UDF in a schema? -> **CREATE FUNCTION**. CREATE FUNCTION on the schema is the object-creation privilege for UDFs.
- **S2-Q8** (E) Why are Snowpark UDFs useful for data engineering teams? -> **They let reusable programmatic logic execute close to Snowflake data**. Snowpark UDFs run reusable code inside Snowflake compute, reducing unnecessary data movement.
- **S2-Q74** (E) Which type of Snowflake function can reach an external service when native execution is not sufficient? -> **External function**. External functions are used to call services outside Snowflake.

- **S2-Q86** (M) A team has reusable Python scoring logic that must be called directly from SQL for each input row. Which Snowflake object is most suitable? -> **Snowpark UDF**. A Snowpark UDF exposes reusable programmatic logic as a SQL-callable function.
- **S2-Q87** (M) A UDF runs correctly but becomes slow after logic is expanded with many complex operations. Which factor from the source is most directly relevant? -> **The complexity of logic inside the UDF**. The source identifies UDF logic complexity as a direct performance factor.
- **S2-Q88** (M) A team wants to reduce overhead when processing batches of values in a Python UDF. Which practice is most aligned with the source? -> **Use vectorized operations where appropriate**. Vectorized processing can improve Snowpark UDF efficiency for batch-style operations.
- **S2-Q149** (M) A SQL UDF is being considered for a workflow that needs loops and branching. What is the better choice? -> **Use a stored procedure instead**. The source states SQL UDFs do not support procedural control flow like stored procedures.
- **S2-Q150** (M) A developer needs a reusable expression that returns a computed value but should not perform multi-step orchestration. Which object fits best? -> **UDF**. A UDF is appropriate for reusable value-returning logic.
- **S2-Q151** (M) A UDF must produce the same result whenever given the same input values. Which characteristic describes this? -> **Deterministic behavior**. Deterministic logic returns the same output for the same input.
- **S2-Q152** (M) A company wants an external service to be called from Snowflake logic. Which object category from the source is relevant? -> **External function**. External functions are used when Snowflake needs to call an external service.
- **S2-Q153** (M) A team is choosing between a UDF and a stored procedure. The requirement is to automate a sequence of SQL steps and control flow. Which should they choose? -> **Stored procedure**. Stored procedures are designed for multi-step procedural workflows.
- **S2-Q154** (M) A team is choosing between a UDF and a stored procedure. The requirement is a reusable calculation in a SELECT list. Which should they choose? -> **UDF**. UDFs are appropriate for reusable value-returning calculations in queries.
- **S2-Q162** (H) A Python UDF is called millions of times and each call performs expensive row-by-row logic. Which source-supported optimization deserves investigation first? -> **Simplify logic and use vectorized processing where possible**. UDF complexity matters, and vectorized operations can reduce overhead for suitable workloads.
- **S2-Q163** (H) A developer proposes a SQL UDF for a process that needs loops, conditional branching, and multiple DML steps. What is the strongest design objection? -> **SQL UDFs are not intended for procedural control-flow workflows**. The source explicitly distinguishes UDFs from stored procedures for procedural logic.
- **S2-Q164** (H) A team chooses a stored procedure for a simple per-row calculation used in many SELECT statements. Which alternative is more appropriate? -> **A scalar UDF**. A scalar UDF is a better fit for reusable per-row value-returning logic.

- **S2-Q192** (H) An engineer uses an external function for logic that could execute natively as a simple Snowflake UDF. What architectural downside should be considered? -> **External calls can add dependency and integration overhead**. External service calls introduce integration and runtime dependencies compared with native execution.
- **S2-Q193** (H) A team needs one reusable function to return a scalar value and another routine to orchestrate multiple statements with conditional branches. Which pairing is correct? -> **Scalar UDF for the calculation; stored procedure for orchestration**. UDFs fit reusable calculations; stored procedures fit multi-step procedural workflows.

X1. Tables, Time Travel, Data Load & Unload, Sharing

Not an MCQ row in the exam, but part of the Coding sections. Bank: ***56 questions*** *(Set 1: 50, Set 2: 6; Easy 20, Medium 20, Hard 16).*

Memory hooks

- **Time Travel** (`AT | BEFORE`, within retention) = see/restore earlier data after a mistake. **Zero-copy cloning** = fast dev copy, shares storage until you change it. **Secure Data Sharing** = live data to another account, **no copy**. **Secure View** = hides the view definition.
- **Loading flow:** `PUT` **(upload local file to internal stage) →** `LIST @stage` **(see files) →** `COPY INTO` **(bulk load).**
- COPY options: `VALIDATION_MODE='RETURN_ERRORS'` = test before loading; `ON_ERROR` = stop/continue/skip; `SKIP_HEADER` = ignore header rows; `FILE_FORMAT` = parsing rules; `FORCE=TRUE` = reload an already-loaded file; **load history** = what was loaded (prevents duplicates).
- Stages: **internal** (Snowflake-managed), **external** (S3/Azure Blob/GCS), `@%table_name` = a table's own stage. S3 access needs an **IAM role with GET and LIST** permissions.

Answer sheet (20 distinct answers, most-asked first)

Answer	Asked	Trigger (what the question is really asking)
Time Travel	4	query or restore historical table data within the retention period / recover from an accidental data change detected soon afterward
VALIDATION_MODE = 'RETURN_ERRORS'	3	test data quality before performing the actual load / validate staged data and return loading errors without inserting rows

Answer	Asked	Trigger (what the question is really asking)
Secure Data Sharing	3	share live Snowflake data with another Snowflake account without copying it / let a partner query provider-controlled data without receiving a physical copy
ON_ERROR	3	choose whether COPY stops, continues, or skips files when errors occur / control how COPY INTO handles data-loading errors
Secure View	3	share selected data while reducing visibility into the underlying view definition / protect a view definition while exposing controlled query results
SKIP_HEADER	3	ignore one or more leading rows in delimited source files / avoid treating a CSV header row as table data
FILE_FORMAT	3	apply delimiter, header, compression, and related parsing settings consistently / define how staged source files should be parsed
External Stage	3	reuse an S3, Azure Blob, or GCS location for data loading / reference files stored in external cloud storage
IAM role with GET and LIST permissions	3	permit Snowpipe or COPY operations to discover and read source objects securely / allow Snowflake to read objects from an external AWS S3 stage
@%table_name	3	address the built-in table stage for a specific table / reference the internal table stage associated with a table
FORCE = TRUE	3	intentionally process a previously loaded file again / reload files even when Snowflake load history says they were already processed
LIST @stage_name	3	display files currently present in a stage / verify which staged objects are available before loading
COPY INTO	3	bulk load staged data into a Snowflake table / load one or more staged files into a target table

Answer	Asked	Trigger (what the question is really asking)
Internal Stage	3	stage files without managing the underlying cloud storage location yourself / store staged files in Snowflake-managed storage
Load History	3	track files that have already been loaded and review load outcomes / confirm prior COPY activity and reduce accidental duplicate ingestion
Zero-copy cloning	3	Which capability lets a development team create a writable environment without initially copying all source storage? / A development team wants a quick copy of production for testing without initially duplicating all data blocks. Which feature should they us…
AT | BEFORE	2	select a prior table state using Snowflake Time Travel syntax / query historical data at or before a specified Time Travel point
PUT	2	upload local files to a Snowflake internal stage / place a local file into an internal staging area before loading
Zero-Copy Cloning	2	create a development copy quickly with minimal initial storage overhead / create a writable logical clone without initially duplicating all underlying storage
Time Travel within the configured retention period	1	A developer accidentally overwrites a table at 10:00 AM and the issue is noticed at noon. No external backup is available. Which Snowflake…

All 56 questions with answers

- **S1-Q1** (M) The data team at a manufacturing company must test data quality before performing the actual load. Which Snowflake feature or command should they select? -> **VALIDATION_MODE = 'RETURN_ERRORS'**. VALIDATION_MODE can inspect load errors without loading the data.
- **S1-Q2** (E) A trainee needs to identify the Snowflake feature that can share live Snowflake data with another Snowflake account without copying it. Which choice is correct? -> **Secure Data Sharing**. Secure Data Sharing exposes live data without copying or exporting it.
- **S1-Q4** (H) During a design review for a banking solution, the team must ensure it can let a partner query provider-controlled data without receiving a physical copy. Which Snowflake

option is the strongest architectural fit? -> **Secure Data Sharing**. Secure Data Sharing exposes live data without copying or exporting it.

- **S1-Q9** (M) In a manufacturing workload, engineers want to select a prior table state using Snowflake Time Travel syntax. What should they use? -> **AT | BEFORE**. AT or BEFORE clauses are used for historical Time Travel queries.
- **S1-Q12** (M) In a insurance workload, engineers want to choose whether COPY stops, continues, or skips files when errors occur. What should they use? -> **ON_ERROR**. ON_ERROR controls COPY INTO behavior when loading errors are encountered.
- **S1-Q17** (H) A banking team is troubleshooting a design. The required outcome is to share selected data while reducing visibility into the underlying view definition, but the current approach does not provide that behavior. Which option should replace the current approach? -> **Secure View**. Secure views help protect view definitions while exposing governed results.
- **S1-Q19** (E) Which Snowflake concept is primarily used to query or restore historical table data within the retention period? -> **Time Travel**. Time Travel provides historical access and recovery within configured retention.
- **S1-Q21** (E) Which option best matches this Snowflake responsibility: ignore one or more leading rows in delimited source files? -> **SKIP_HEADER**. SKIP_HEADER specifies how many initial rows to ignore.
- **S1-Q22** (H) A e-commerce team is troubleshooting a design. The required outcome is to apply delimiter, header, compression, and related parsing settings consistently, but the current approach does not provide that behavior. Which option should replace the current approach? -> **FILE_FORMAT**. FILE_FORMAT supplies parsing rules for staged files.
- **S1-Q23** (E) For the requirement "validate staged data and return loading errors without inserting rows," which Snowflake term is the best fit? -> **VALIDATION_MODE = 'RETURN_ERRORS'**. VALIDATION_MODE can inspect load errors without loading the data.
- **S1-Q28** (M) In a education workload, engineers want to let a partner query provider-controlled data without receiving a physical copy. What should they use? -> **Secure Data Sharing**. Secure Data Sharing exposes live data without copying or exporting it.
- **S1-Q32** (M) A media team needs to reuse an S3, Azure Blob, or GCS location for data loading. Which Snowflake option best meets this requirement? -> **External Stage**. An external stage references files in supported external cloud storage.
- **S1-Q34** (E) For the requirement "define how staged source files should be parsed," which Snowflake term is the best fit? -> **FILE_FORMAT**. FILE_FORMAT supplies parsing rules for staged files.
- **S1-Q35** (H) A retail platform must permit Snowpipe or COPY operations to discover and read source objects securely. The team wants the most direct Snowflake mechanism rather than a workaround. Which choice is best? -> **IAM role with GET and LIST permissions**. For S3 access, Snowflake needs permissions that allow listing and reading the required objects.
- **S1-Q43** (H) A financial services platform must address the built-in table stage for a specific table. The team wants the most direct Snowflake mechanism rather than a workaround.

Which choice is best? -> **@%table_name**. The @%table_name syntax refers to a table stage.

- **S1-Q47** (E) A trainee needs to identify the Snowflake feature that can upload local files to a Snowflake internal stage. Which choice is correct? -> **PUT**. PUT uploads local files to internal stages.
- **S1-Q48** (E) For the requirement "reference the internal table stage associated with a table," which Snowflake term is the best fit? -> **@%table_name**. The @%table_name syntax refers to a table stage.
- **S1-Q49** (M) The data team at a digital payments company must share selected data while reducing visibility into the underlying view definition. Which Snowflake feature or command should they select? -> **Secure View**. Secure views help protect view definitions while exposing governed results.
- **S1-Q50** (H) A telecommunications team is troubleshooting a design. The required outcome is to intentionally process a previously loaded file again, but the current approach does not provide that behavior. Which option should replace the current approach? -> **FORCE = TRUE**. FORCE=TRUE overrides normal duplicate-file load-history protection.
- **S1-Q57** (E) For the requirement "display files currently present in a stage," which Snowflake term is the best fit? -> **LIST @stage_name**. LIST shows files available in a Snowflake stage.
- **S1-Q60** (E) Which Snowflake capability directly supports the need to bulk load staged data into a Snowflake table? -> **COPY INTO**. COPY INTO is Snowflake's primary bulk loading command from a stage into a table.
- **S1-Q63** (M) The data team at a banking company must permit Snowpipe or COPY operations to discover and read source objects securely. Which Snowflake feature or command should they select? -> **IAM role with GET and LIST permissions**. For S3 access, Snowflake needs permissions that allow listing and reading the required objects.
- **S1-Q67** (M) A media team needs to recover from an accidental data change detected soon afterward. Which Snowflake option best meets this requirement? -> **Time Travel**. Time Travel provides historical access and recovery within configured retention.
- **S1-Q68** (M) A manufacturing platform is being redesigned so it can create a development copy quickly with minimal initial storage overhead. Which choice is most appropriate? -> **Zero-Copy Cloning**. Zero-copy cloning creates metadata-based clones that share existing storage until changes occur.
- **S1-Q74** (M) A financial services team needs to apply delimiter, header, compression, and related parsing settings consistently. Which Snowflake option best meets this requirement? -> **FILE_FORMAT**. FILE_FORMAT supplies parsing rules for staged files.
- **S1-Q76** (H) A education engineer is asked to meet this constraint: stage files without managing the underlying cloud storage location yourself. Which option addresses the constraint with the least unnecessary complexity? -> **Internal Stage**. Snowflake manages storage for internal stages.
- **S1-Q77** (M) The data team at a marketing company must address the built-in table stage for a specific table. Which Snowflake feature or command should they select? -> **@%table_name**.

The @%table_name syntax refers to a table stage.

- **S1-Q78** (E) For the requirement "query historical data at or before a specified Time Travel point," which Snowflake term is the best fit? -> **AT | BEFORE**. AT or BEFORE clauses are used for historical Time Travel queries.
- **S1-Q83** (M) The data team at a media company must verify which staged objects are available before loading. Which Snowflake feature or command should they select? -> **LIST @stage_name**. LIST shows files available in a Snowflake stage.
- **S1-Q85** (H) During a design review for a e-commerce solution, the team must ensure it can reuse an S3, Azure Blob, or GCS location for data loading. Which Snowflake option is the strongest architectural fit? -> **External Stage**. An external stage references files in supported external cloud storage.
- **S1-Q96** (H) A financial services platform must test data quality before performing the actual load. The team wants the most direct Snowflake mechanism rather than a workaround. Which choice is best? -> **VALIDATION_MODE = 'RETURN_ERRORS'**. VALIDATION_MODE can inspect load errors without loading the data.
- **S1-Q99** (E) A trainee needs to identify the Snowflake feature that can allow Snowflake to read objects from an external AWS S3 stage. Which choice is correct? -> **IAM role with GET and LIST permissions**. For S3 access, Snowflake needs permissions that allow listing and reading the required objects.
- **S1-Q102** (E) Which Snowflake concept is primarily used to track files that have already been loaded and review load outcomes? -> **Load History**. Snowflake maintains load-history metadata for processed files and load results.
- **S1-Q103** (M) A insurance team needs to avoid treating a CSV header row as table data. Which Snowflake option best meets this requirement? -> **SKIP_HEADER**. SKIP_HEADER specifies how many initial rows to ignore.
- **S1-Q105** (H) A telecommunications engineer is asked to meet this constraint: choose whether COPY stops, continues, or skips files when errors occur. Which option addresses the constraint with the least unnecessary complexity? -> **ON_ERROR**. ON_ERROR controls COPY INTO behavior when loading errors are encountered.
- **S1-Q108** (E) For the requirement "store staged files in Snowflake-managed storage," which Snowflake term is the best fit? -> **Internal Stage**. Snowflake manages storage for internal stages.
- **S1-Q114** (E) A trainee needs to identify the Snowflake feature that can reload files even when Snowflake load history says they were already processed. Which choice is correct? -> **FORCE = TRUE**. FORCE=TRUE overrides normal duplicate-file load-history protection.
- **S1-Q118** (M) A telecommunications team needs to intentionally process a previously loaded file again. Which Snowflake option best meets this requirement? -> **FORCE = TRUE**. FORCE=TRUE overrides normal duplicate-file load-history protection.
- **S1-Q120** (H) A travel team is troubleshooting a design. The required outcome is to confirm prior COPY activity and reduce accidental duplicate ingestion, but the current approach does

not provide that behavior. Which option should replace the current approach? -> **Load History**. Snowflake maintains load-history metadata for processed files and load results.

- **S1-Q129** (E) Which option best matches this Snowflake responsibility: protect a view definition while exposing controlled query results? -> **Secure View**. Secure views help protect view definitions while exposing governed results.
- **S1-Q137** (H) A retail team is troubleshooting a design. The required outcome is to load one or more staged files into a target table, but the current approach does not provide that behavior. Which option should replace the current approach? -> **COPY INTO**. COPY INTO is Snowflake's primary bulk loading command from a stage into a table.
- **S1-Q153** (M) An architect for a digital payments system has one key requirement: stage files without managing the underlying cloud storage location yourself. Which option best satisfies it? -> **Internal Stage**. Snowflake manages storage for internal stages.
- **S1-Q157** (E) A trainee needs to identify the Snowflake feature that can reference files stored in external cloud storage. Which choice is correct? -> **External Stage**. An external stage references files in supported external cloud storage.
- **S1-Q159** (E) For the requirement "create a writable logical clone without initially duplicating all underlying storage," which Snowflake term is the best fit? -> **Zero-Copy Cloning**. Zero-copy cloning creates metadata-based clones that share existing storage until changes occur.
- **S1-Q167** (M) The data team at a supply chain company must load one or more staged files into a target table. Which Snowflake feature or command should they select? -> **COPY INTO**. COPY INTO is Snowflake's primary bulk loading command from a stage into a table.
- **S1-Q168** (H) A logistics team is troubleshooting a design. The required outcome is to avoid treating a CSV header row as table data, but the current approach does not provide that behavior. Which option should replace the current approach? -> **SKIP_HEADER**. SKIP_HEADER specifies how many initial rows to ignore.
- **S1-Q170** (E) Which Snowflake capability directly supports the need to control how COPY INTO handles data-loading errors? -> **ON_ERROR**. ON_ERROR controls COPY INTO behavior when loading errors are encountered.
- **S1-Q178** (H) During a design review for a healthcare solution, the team must ensure it can verify which staged objects are available before loading. Which Snowflake option is the strongest architectural fit? -> **LIST @stage_name**. LIST shows files available in a Snowflake stage.
- **S1-Q187** (M) A e-commerce platform is being redesigned so it can confirm prior COPY activity and reduce accidental duplicate ingestion. Which choice is most appropriate? -> **Load History**. Snowflake maintains load-history metadata for processed files and load results.
- **S1-Q197** (M) The data team at a manufacturing company must place a local file into an internal staging area before loading. Which Snowflake feature or command should they select? -> **PUT**. PUT uploads local files to internal stages.
- **S2-Q65** (E) Which capability lets a development team create a writable environment without initially copying all source storage? -> **Zero-copy cloning**. Cloning initially shares underlying

storage while allowing independent changes.

- **S2-Q66** (E) Which Snowflake feature allows access to earlier versions of table data after an accidental change? -> **Time Travel**. Time Travel provides access to historical table states within retention.
- **S2-Q131** (M) A development team wants a quick copy of production for testing without initially duplicating all data blocks. Which feature should they use? -> **Zero-copy cloning**. Zero-copy cloning creates a fast writable copy with shared storage initially.
- **S2-Q132** (M) An analyst accidentally updates a table and needs to query it as it existed earlier in the day. Which feature should be used? -> **Time Travel**. Time Travel provides access to historical table versions within retention.
- **S2-Q172** (H) A company wants development copies of production with minimal initial storage overhead but needs developers to change data independently afterward. Which feature best satisfies both needs? -> **Zero-copy cloning**. Zero-copy cloning shares existing storage initially while allowing later independent changes.
- **S2-Q173** (H) A developer accidentally overwrites a table at 10:00 AM and the issue is noticed at noon. No external backup is available. Which Snowflake capability should be evaluated first? -> **Time Travel within the configured retention period**. Time Travel is designed for accessing earlier table states within retention.

X2. Semi-structured Data / Cortex

Not an MCQ row in the exam, but part of the Coding sections. Bank: ***25 questions*** *(Set 1: 6, Set 2: 19; Easy 9, Medium 9, Hard 7).*

Memory hooks

- **VARIANT** = the type for semi-structured data (JSON); query nested fields directly. Hot JSON fields slow? **Extract them into their own columns.**
- **Cortex:** built-in AI functions = use AI from **SQL** (summarize/classify); **Cortex Search** = semantic (meaning-based) retrieval; **Cortex Agents** = natural-language assistant over governed data; **Snowpark ML** = custom Python model.
- AI features still follow **RBAC** on the underlying governed data.

Answer sheet (15 distinct answers, most-asked first)

Answer	Asked	Trigger (what the question is really asking)
VARIANT	5	store semi-structured values such as nested JSON / land evolving JSON structures before extracting selected attributes
Nested field access from VARIANT	3	query individual attributes inside semi-structured JSON values / analyze selected JSON attributes without first redesigning the data into fixed relational columns
Cortex Search	3	Which capability is associated with semantic retrieval of relevant information? / A company wants users to retrieve relevant passages from a large knowledge corpus using semantic meaning rather than exact keywords. Which…
Snowpark ML	2	Which technology is more appropriate for custom Python-based model development mentioned in the source? / A data science team wants to build and run a custom Python model while keeping training data inside Snowflake. Which capability is the bett…
Cortex Agents	2	A user asks natural-language questions over governed Snowflake data and expects AI-driven interaction. Which capability is most closely ali… / A company wants a natural-language assistant that can interact with governed Snowflake data using native AI services. Which capability is t…
SQL	1	Snowflake built-in AI functions are mainly intended to let users access AI capabilities through which familiar interface?
Summarization or classification	1	Which type of task is most aligned with text inference capabilities in Snowflake?
AI-driven interaction with Snowflake data	1	What is the central purpose of Cortex Agents in the source material?
Governed access to data without	1	Which phrase best matches a benefit of keeping AI processing in Snowflake?

Answer	Asked	Trigger (what the question is really asking)
unnecessary movement		
Snowflake built-in AI text functions	1	A company wants a SQL-accessible AI capability to summarize support tickets stored in Snowflake. Which family of capability is appropriate?
Use Snowflake AI capabilities together with RBAC	1	A company wants AI-generated insights but must keep authorization governed by existing Snowflake roles. Which approach aligns with the sour…
Extract frequently queried attributes into separate columns	1	A team queries two JSON attributes constantly and performance is poor. Which source-supported optimization approach is reasonable?
Materialize those frequently queried attributes into dedicated columns	1	An ingestion team loads evolving JSON and later learns that two nested fields drive 90% of filters. Which redesign most directly targets qu…
Use Snowpark ML for the custom model workflow	1	A team uses a generative AI function for a use case that actually requires training and deploying a custom predictive model. Which correcti…
Apply Snowflake RBAC to the underlying governed data and app access	1	An AI feature must respect different access levels for analysts and auditors. Which design principle should remain in force?

All 25 questions with answers

- **S1-Q37** (E) For the requirement "store semi-structured values such as nested JSON," which Snowflake term is the best fit? -> **VARIANT**. VARIANT is designed to store and query semi-structured data.
- **S1-Q40** (M) A banking platform is being redesigned so it can land evolving JSON structures before extracting selected attributes. Which choice is most appropriate? -> **VARIANT**. VARIANT is designed to store and query semi-structured data.
- **S1-Q45** (H) A manufacturing engineer is asked to meet this constraint: land evolving JSON structures before extracting selected attributes. Which option addresses the constraint with the least unnecessary complexity? -> **VARIANT**. VARIANT is designed to store and query semi-structured data.

- **S1-Q53** (E) For the requirement "query individual attributes inside semi-structured JSON values," which Snowflake term is the best fit? -> **Nested field access from VARIANT**. Snowflake supports SQL access to nested elements stored in VARIANT values.
- **S1-Q66** (M) The data team at a healthcare company must analyze selected JSON attributes without first redesigning the data into fixed relational columns. Which Snowflake feature or command should they select? -> **Nested field access from VARIANT**. Snowflake supports SQL access to nested elements stored in VARIANT values.
- **S1-Q164** (H) A financial services platform must analyze selected JSON attributes without first redesigning the data into fixed relational columns. The team wants the most direct Snowflake mechanism rather than a workaround. Which choice is best? -> **Nested field access from VARIANT**. Snowflake supports SQL access to nested elements stored in VARIANT values.
- **S2-Q25** (E) Snowflake built-in AI functions are mainly intended to let users access AI capabilities through which familiar interface? -> **SQL**. The source emphasizes Snowflake AI capabilities integrated into SQL workflows.
- **S2-Q26** (E) Which type of task is most aligned with text inference capabilities in Snowflake? -> **Summarization or classification**. Text inference commonly supports generation, summarization, and classification.
- **S2-Q27** (E) What is the central purpose of Cortex Agents in the source material? -> **AI-driven interaction with Snowflake data**. Cortex Agents provide an AI-oriented interface for working with Snowflake data.
- **S2-Q28** (E) Which capability is associated with semantic retrieval of relevant information? -> **Cortex Search**. Cortex Search is intended for AI-powered semantic retrieval.
- **S2-Q29** (E) Which phrase best matches a benefit of keeping AI processing in Snowflake? -> **Governed access to data without unnecessary movement**. Native AI capabilities can work with governed Snowflake data while reducing data movement.
- **S2-Q30** (E) Which technology is more appropriate for custom Python-based model development mentioned in the source? -> **Snowpark ML**. The source distinguishes custom Python ML workflows from generic generative AI use cases.
- **S2-Q67** (E) Which data type is designed for semi-structured records such as evolving JSON payloads? -> **VARIANT**. VARIANT stores semi-structured data such as JSON.
- **S2-Q120** (M) A company wants a SQL-accessible AI capability to summarize support tickets stored in Snowflake. Which family of capability is appropriate? -> **Snowflake built-in AI text functions**. The source positions built-in AI functions for text generation, summarization, and classification via SQL.
- **S2-Q121** (M) A data science team wants to build and run a custom Python model while keeping training data inside Snowflake. Which capability is the better fit? -> **Snowpark ML**. Snowpark ML is the source-supported path for custom Python-based model development.
- **S2-Q122** (M) A user asks natural-language questions over governed Snowflake data and expects AI-driven interaction. Which capability is most closely aligned? -> **Cortex Agents**. Cortex Agents support AI-powered interaction with Snowflake data.

- **S2-Q123** (M) A company wants users to retrieve relevant passages from a large knowledge corpus using semantic meaning rather than exact keywords. Which capability fits? -> **Cortex Search**. Cortex Search focuses on semantic/AI-powered retrieval.
- **S2-Q124** (M) A company wants AI-generated insights but must keep authorization governed by existing Snowflake roles. Which approach aligns with the source? -> **Use Snowflake AI capabilities together with RBAC**. Native AI can be combined with RBAC to preserve governed access.
- **S2-Q133** (M) An application emits JSON records whose fields evolve over time. The team wants to load them without defining every nested field as a column first. Which data type fits? -> **VARIANT**. VARIANT is designed for semi-structured data such as JSON.
- **S2-Q134** (M) A team queries two JSON attributes constantly and performance is poor. Which source-supported optimization approach is reasonable? -> **Extract frequently queried attributes into separate columns**. The source suggests extracting frequently accessed semi-structured fields into columns.
- **S2-Q174** (H) An ingestion team loads evolving JSON and later learns that two nested fields drive 90% of filters. Which redesign most directly targets query efficiency? -> **Materialize those frequently queried attributes into dedicated columns**. Extracting heavily queried fields can improve filter efficiency without redesigning the whole dataset.
- **S2-Q175** (H) A knowledge-search application must find semantically relevant text passages, not just exact keyword matches. Which Snowflake AI capability is the better architectural match? -> **Cortex Search**. Cortex Search is designed for semantic retrieval.
- **S2-Q176** (H) A company wants a natural-language assistant that can interact with governed Snowflake data using native AI services. Which capability is the closest match? -> **Cortex Agents**. Cortex Agents provide AI-powered interaction with Snowflake data.
- **S2-Q177** (H) A team uses a generative AI function for a use case that actually requires training and deploying a custom predictive model. Which correction best fits the source? -> **Use Snowpark ML for the custom model workflow**. The source distinguishes custom Python ML development from generic built-in/generative AI functionality.
- **S2-Q178** (H) An AI feature must respect different access levels for analysts and auditors. Which design principle should remain in force? -> **Apply Snowflake RBAC to the underlying governed data and app access**. AI access should remain governed by Snowflake roles and privileges.